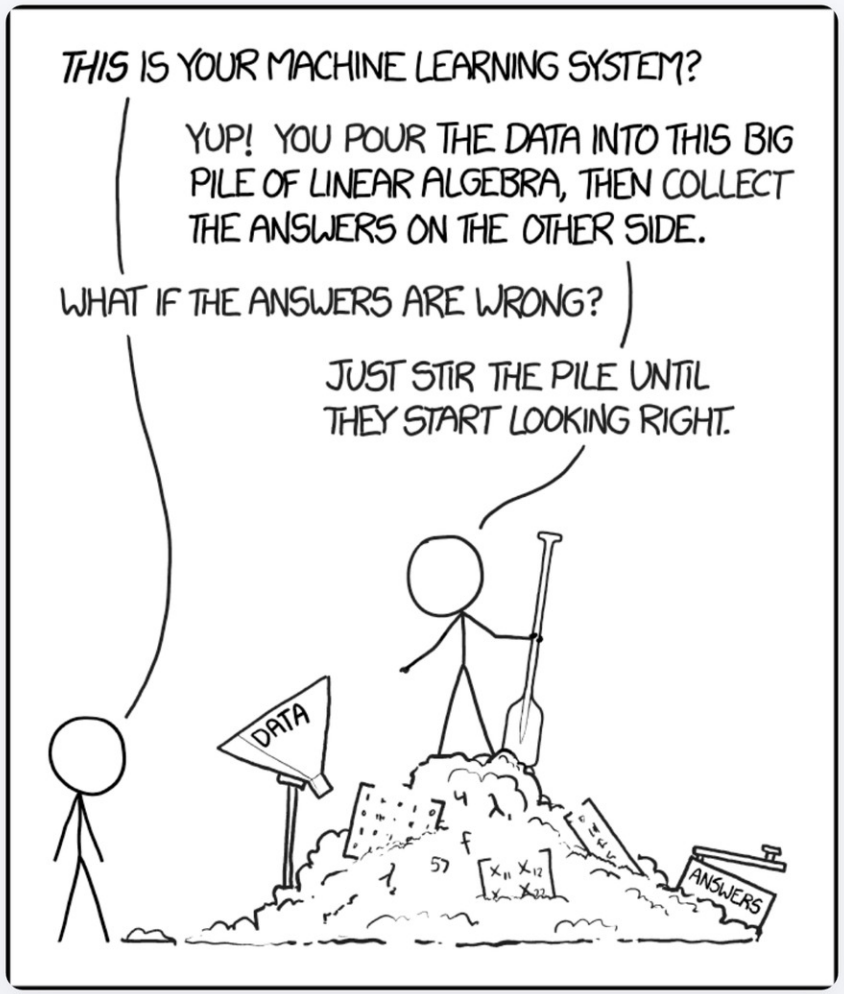


"Machine Learning", xkcd 1838, by Randall Munroe (xkcd.com/1838). CC BY-NC 2.5.

MACHINE LEARNING

Neural networks

Part 2: training a network.

LAST TIME

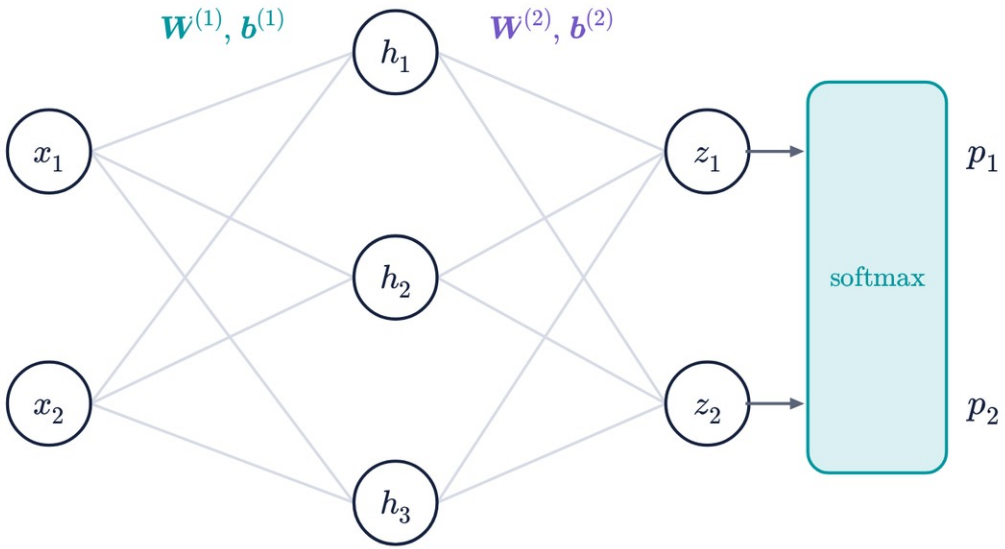
Forward propagation

$$\boldsymbol{h} = \sigma(\boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)})$$
$$\boldsymbol{z} = \boldsymbol{W}^{(2)}\boldsymbol{h} + \boldsymbol{b}^{(2)}$$
$$\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z})$$

- Train by gradient descent:

  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla \mathcal{L}(\boldsymbol{\theta})$

- We need $\partial \mathcal{L} / \partial w$ for every weight w. Where does it come from?



BACKPROPAGATION

The network is one formula

A small network: two inputs, two hidden units, one output. Written out in full:

$$p = \sigma\Big(v_1\,\sigma(w_{11}x_1 + w_{12}x_2 + b_1) + v_2\,\sigma(w_{21}x_1 + w_{22}x_2 + b_2) + c\Big)$$

BACKPROPAGATION

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The loss of one training point, here squared error:

$$\mathcal{L} = \tfrac{1}{2}(p - y)^2$$

BACKPROPAGATION

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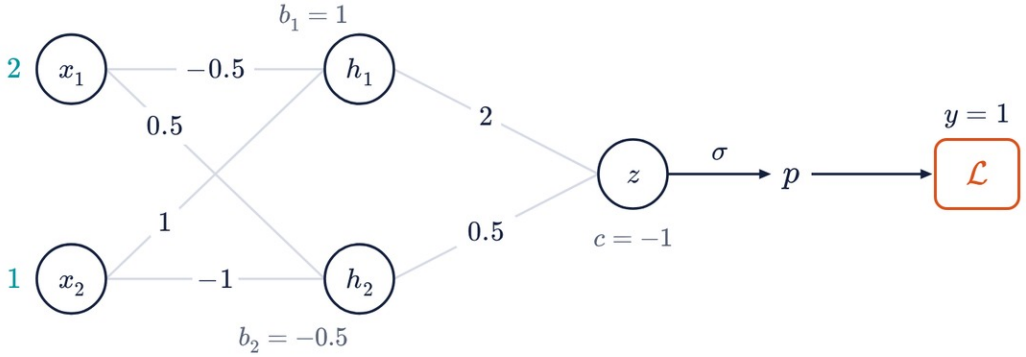
The loss of one training point, here squared error:

$$\mathcal{L} = \tfrac{1}{2}(p - y)^2$$

Nine weights, all inside one nested formula. We need $\partial\mathcal{L}/\partial w$ for each one.

A worked example

- One training point: $\boldsymbol{x} = (2, 1)$, label $y = 1$.
  The weights are on the connections.
  Sigmoid units; squared error $\mathcal{L} = \frac{1}{2}(p - y)^2$.



A worked example

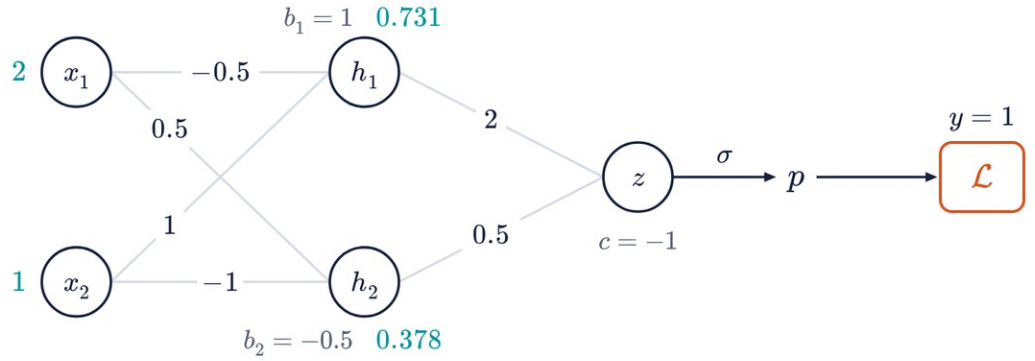
- One training point: $\boldsymbol{x} = (2, 1)$, label $y = 1$.
  The weights are on the connections.
  Sigmoid units; squared error $\mathcal{L} = \frac{1}{2}(p - y)^2$.
- Hidden layer:

  $a_1 = -0.5 \cdot 2 + 1 \cdot 1 + 1 = 1$

  $h_1 = \sigma(1) = 0.731$

  $a_2 = 0.5 \cdot 2 - 1 \cdot 1 - 0.5 = -0.5$

  $h_2 = \sigma(-0.5) = 0.378$



A worked example

- One training point: $\boldsymbol{x} = (2, 1)$, label $y = 1$.
  The weights are on the connections.
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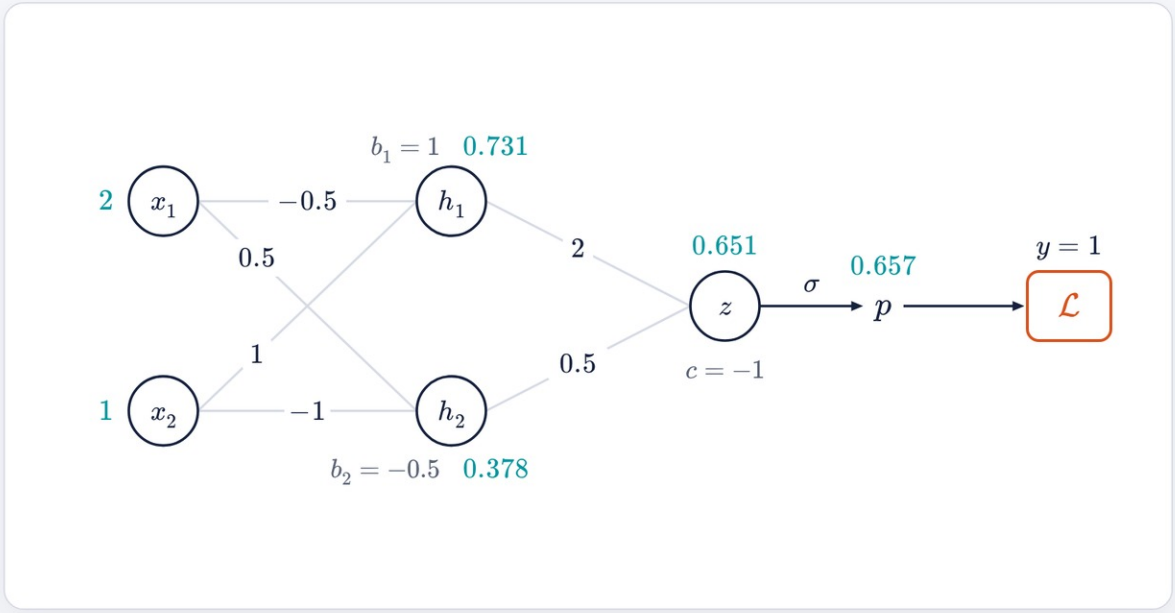
  $h_1 = \sigma(1) = 0.731$

  $a_2 = 0.5 \cdot 2 - 1 \cdot 1 - 0.5 = -0.5$

  $h_2 = \sigma(-0.5) = 0.378$
- Output:

  $z = 2 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = 0.651$

  $p = \sigma(0.651) = 0.657$



A worked example

- One training point: $\boldsymbol{x} = (2, 1)$, label $y = 1$.
  The weights are on the connections.
  Sigmoid units; squared error $\mathcal{L} = \frac{1}{2}(p - y)^2$.
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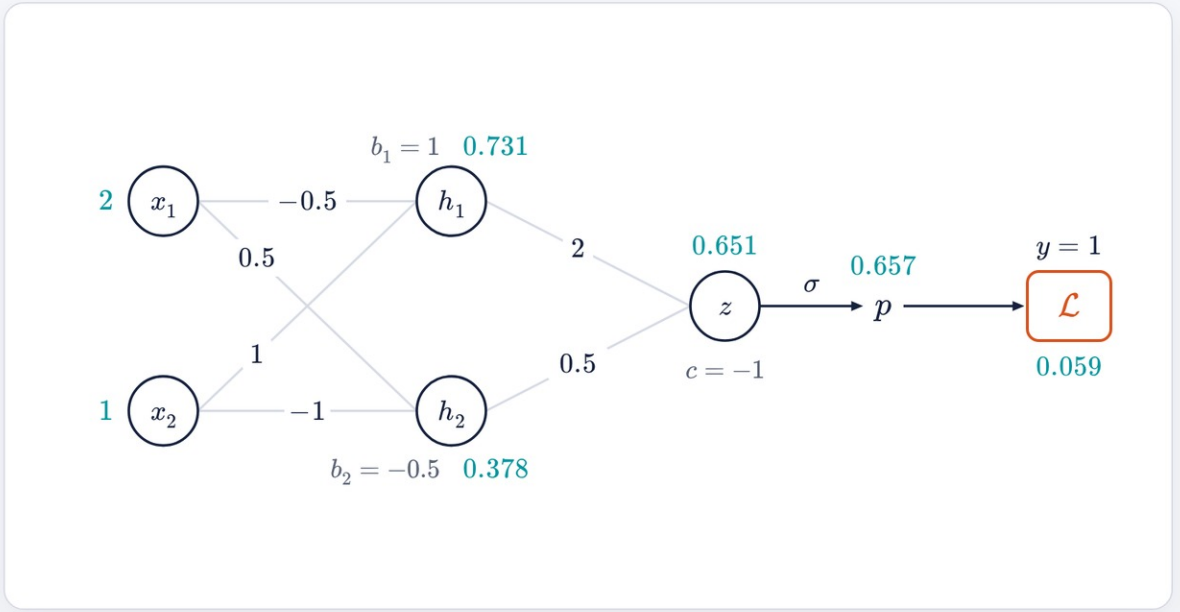
  $a_2 = 0.5 \cdot 2 - 1 \cdot 1 - 0.5 = -0.5$

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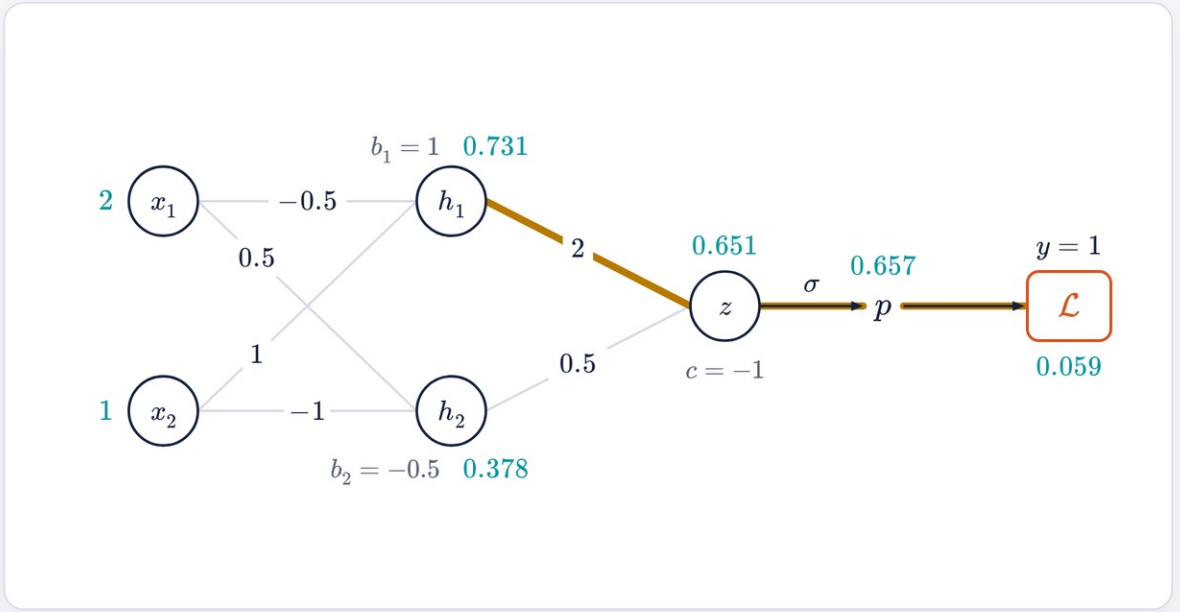
  $p = \sigma(0.651) = 0.657$
- Loss:

  $\mathcal{L} = \frac{1}{2}(0.657 - 1)^2 = 0.059$



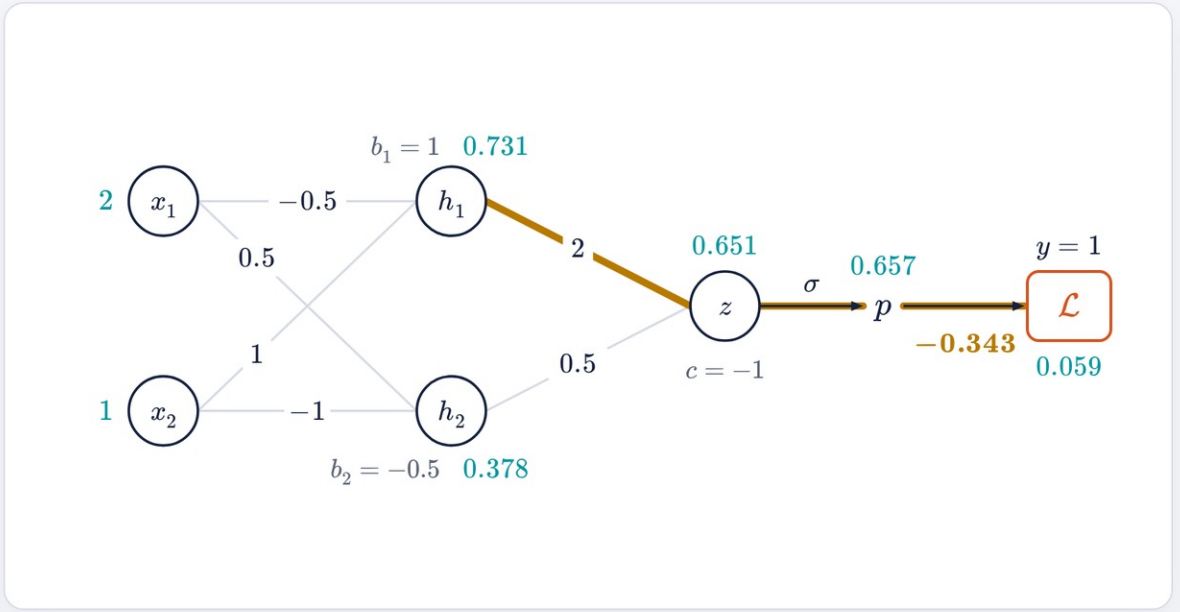
An output weight: v_1

- v_1 changes the loss along the highlighted path, $v_1 \to z \to p \to \mathcal{L}$. The chain rule multiplies the derivatives along it:
$$\frac{\partial \mathcal{L}}{\partial v_1} = \frac{\partial \mathcal{L}}{\partial p} \cdot \frac{\partial p}{\partial z} \cdot \frac{\partial z}{\partial v_1}$$



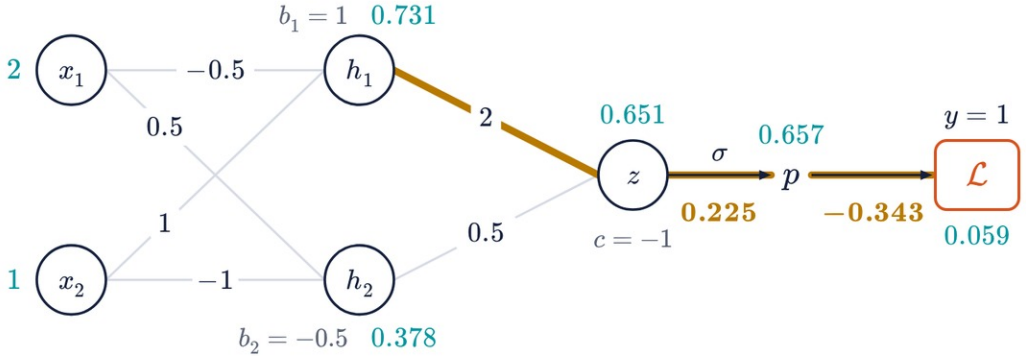
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  $$\frac{\partial \mathcal{L}}{\partial v_1} = \frac{\partial \mathcal{L}}{\partial p} \cdot \frac{\partial p}{\partial z} \cdot \frac{\partial z}{\partial v_1}$$
- $\partial \mathcal{L}/\partial p = p - y = 0.657 - 1 = -0.343$



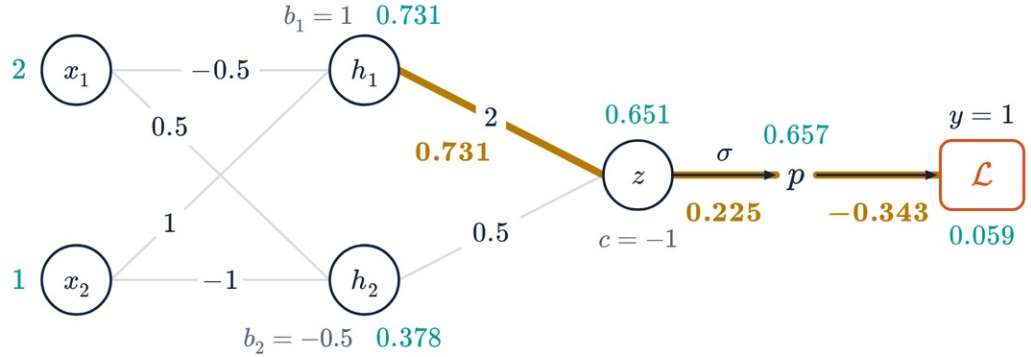
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- $\partial \mathcal{L}/\partial p = p - y = 0.657 - 1 = -0.343$
- $\partial p/\partial z = p(1-p) = 0.225$
  0.657×0.343. The sigmoid's slope.



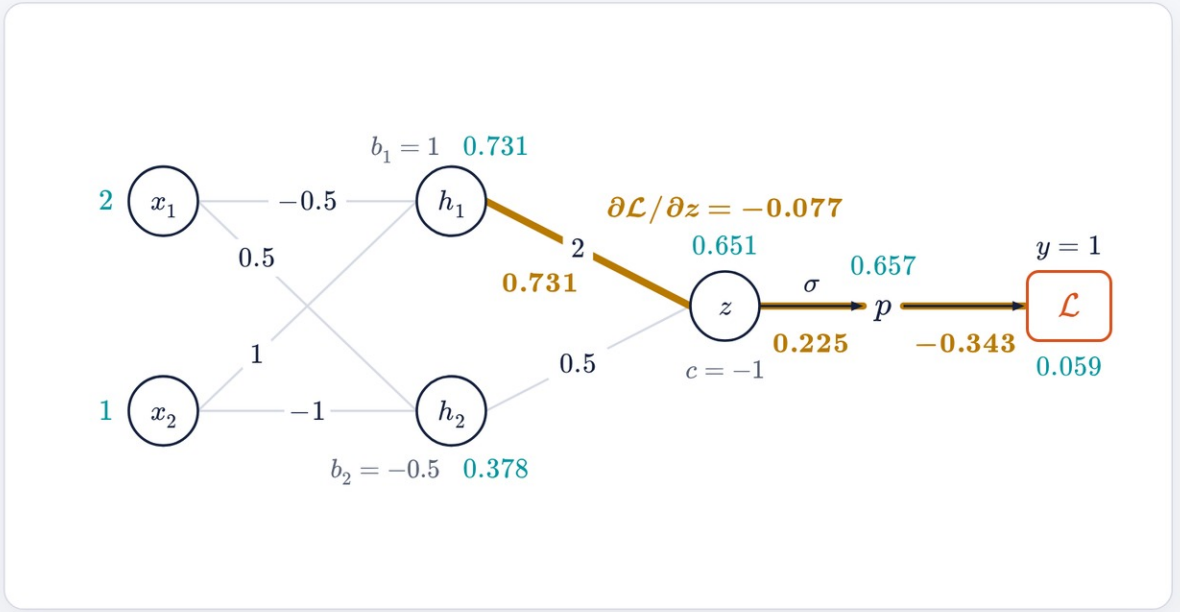
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- $\partial p/\partial z = p(1-p) = 0.225$
  0.657×0.343. The sigmoid's slope.
- $\partial z/\partial v_1 = h_1 = 0.731$
  Because $z = v_1 h_1 + v_2 h_2 + c$.



An output weight: v_1

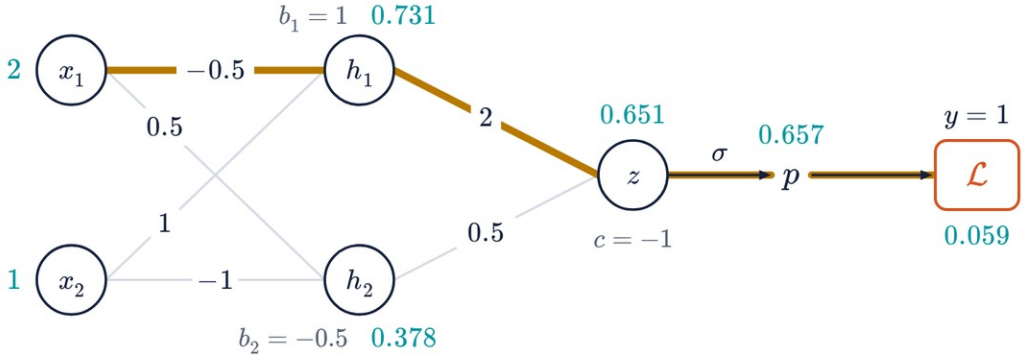
- v_1 changes the loss along the highlighted path, $v_1 \to z \to p \to \mathcal{L}$. The chain rule multiplies the derivatives along it:
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  0.657×0.343. The sigmoid's slope.
- $\partial z/\partial v_1 = h_1 = 0.731$
  Because $z = v_1 h_1 + v_2 h_2 + c$.
- Multiply. Keep the first two factors for later:
  $$\frac{\partial \mathcal{L}}{\partial z} = -0.343 \times 0.225 = -0.077$$
  $$\frac{\partial \mathcal{L}}{\partial v_1} = -0.077 \times 0.731 = -0.056$$



A hidden weight: w_{11}

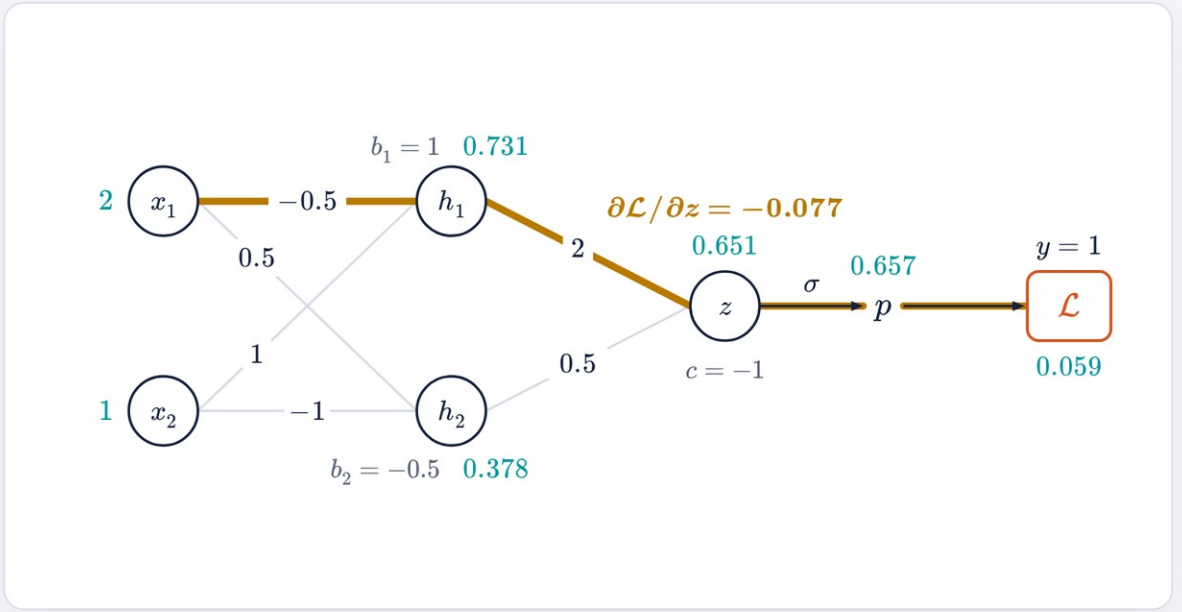
- The path is longer:
  $w_{11} \to a_1 \to h_1 \to z \to p \to \mathcal{L}$.

$$\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial z} \cdot \frac{\partial z}{\partial h_1} \cdot \frac{\partial h_1}{\partial a_1} \cdot \frac{\partial a_1}{\partial w_{11}}$$



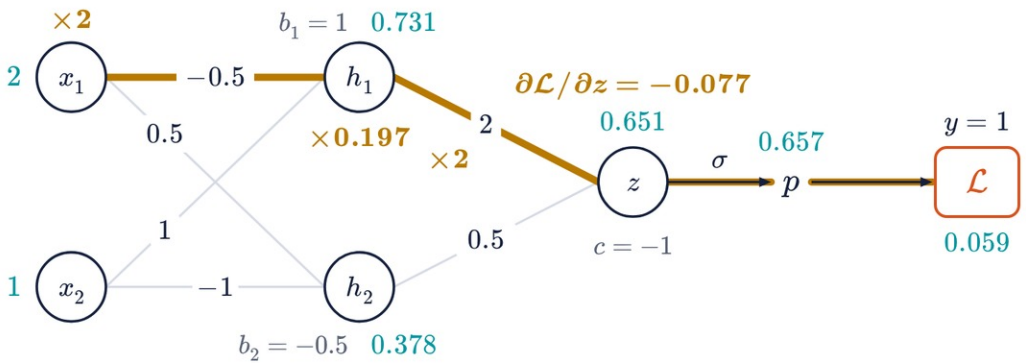
A hidden weight: w_{11}

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- $\partial \mathcal{L}/\partial z = -0.077$, already computed for v_1. Reuse it.



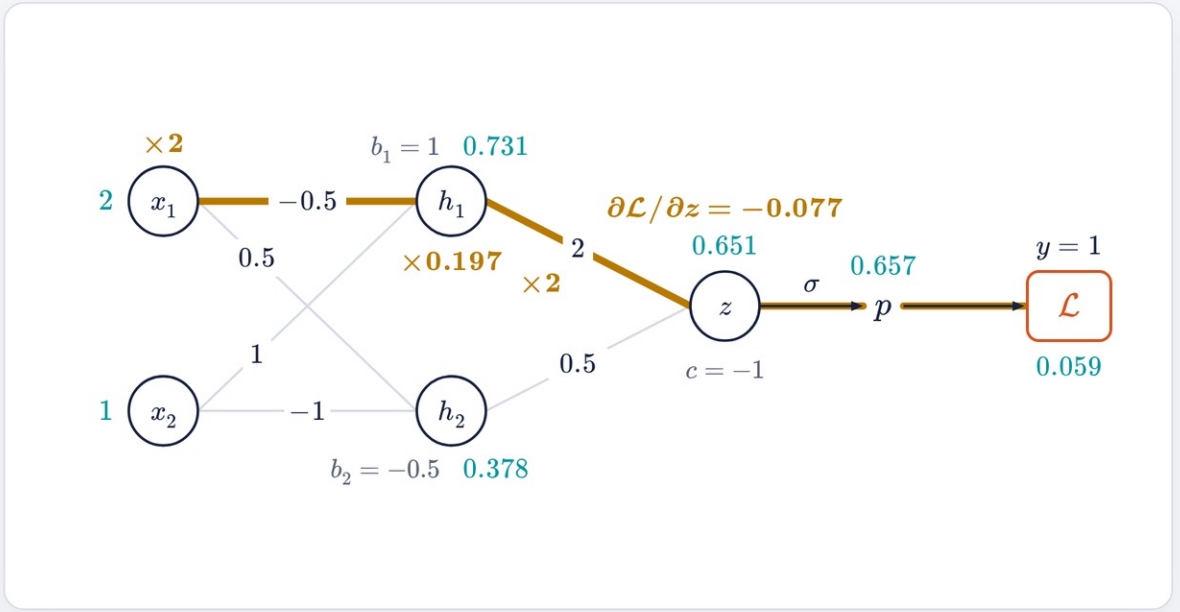
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  $w_{11} \to a_1 \to h_1 \to z \to p \to \mathcal{L}$.
  $$\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial z} \cdot \frac{\partial z}{\partial h_1} \cdot \frac{\partial h_1}{\partial a_1} \cdot \frac{\partial a_1}{\partial w_{11}}$$
- $\partial \mathcal{L}/\partial z = -0.077$, already computed for v_1. Reuse it.
- The new factors:
  $$\partial z/\partial h_1 = v_1 = 2$$
  $$\partial h_1/\partial a_1 = h_1(1 - h_1) = 0.731 \times 0.269 = 0.197$$
  $$\partial a_1/\partial w_{11} = x_1 = 2$$



A hidden weight: w_{11}

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  $$\partial h_1/\partial a_1 = h_1(1 - h_1) = 0.731 \times 0.269 = 0.197$$
  $$\partial a_1/\partial w_{11} = x_1 = 2$$
- Multiply:
  $$\frac{\partial \mathcal{L}}{\partial w_{11}} = -0.077 \times 2 \times 0.197 \times 2 = -0.061$$
  With another activation φ, use $\varphi'(a_1)$ in place of $h_1(1 - h_1)$.

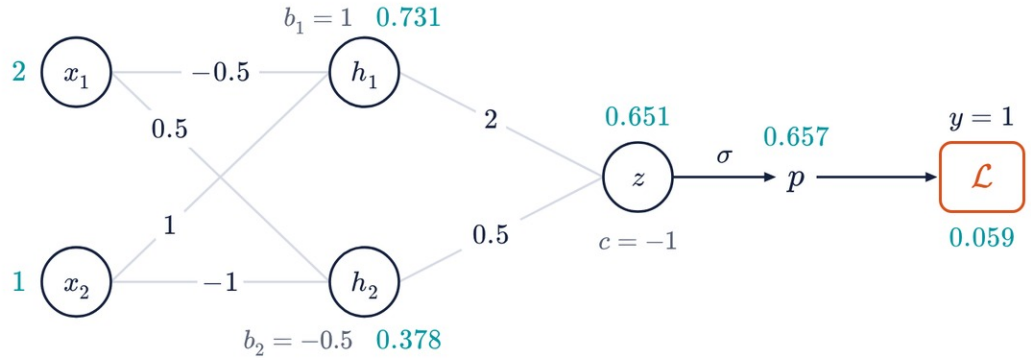


BACKPROPAGATION

Update the weights

- Gradient descent, with learning rate $\eta = 1$:

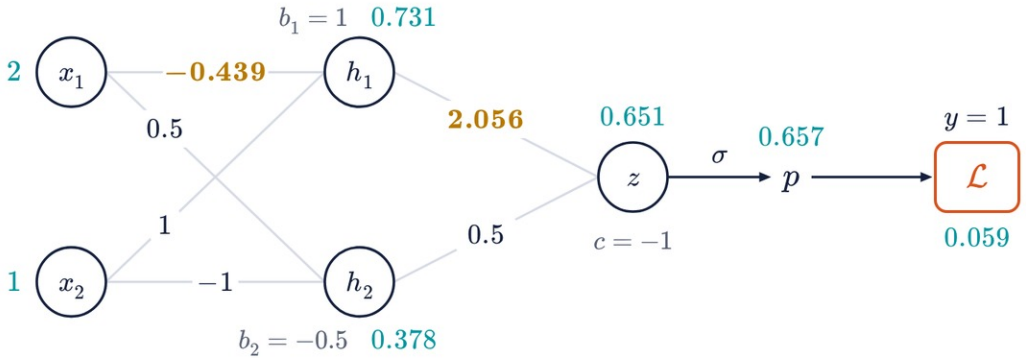
$$w \leftarrow w - \eta \frac{\partial \mathcal{L}}{\partial w}$$



BACKPROPAGATION

Update the weights

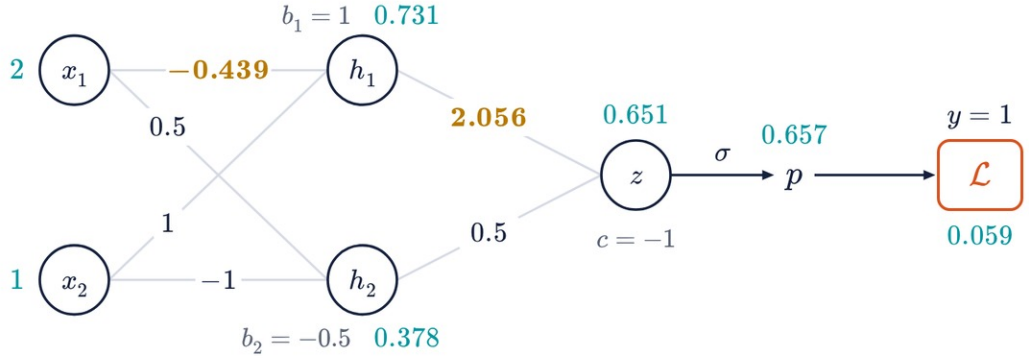
- Gradient descent, with learning rate $\eta = 1$:
  $w \leftarrow w - \eta \dfrac{\partial \mathcal{L}}{\partial w}$
- $v_1 \leftarrow 2 - (-0.056) = 2.056$
  $w_{11} \leftarrow -0.5 - (-0.061) = -0.439$



BACKPROPAGATION

Update the weights

- Gradient descent, with learning rate $\eta = 1$:
  $w \leftarrow w - \eta \dfrac{\partial \mathcal{L}}{\partial w}$
- $v_1 \leftarrow 2 - (-0.056) = 2.056$
  $w_{11} \leftarrow -0.5 - (-0.061) = -0.439$
- Do the same for all nine weights, with every gradient computed from the old weights. Then the next training point.



BACKPROPAGATION

Now with cross-entropy

- Same network, same point, same forward pass: $p = 0.657$. Only the loss changes:

$$\begin{aligned}\mathcal{L} &= -y \log p - (1-y)\log(1-p) \\ &= -\log 0.657 = 0.420\end{aligned}$$

	chain	squared error
$\partial \mathcal{L}/\partial z$		-0.077
$\partial \mathcal{L}/\partial v_1$	$\partial \mathcal{L}/\partial z \times 0.731$	-0.056
$\partial \mathcal{L}/\partial w_{11}$	$\partial \mathcal{L}/\partial z \times 2 \times 0.197 \times 2$	-0.061

BACKPROPAGATION

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$$\begin{aligned}\mathcal{L} &= -y \log p - (1-y)\log(1-p) \\ &= -\log 0.657 = 0.420\end{aligned}$$

- So only the first factor changes:

$$\frac{\partial \mathcal{L}}{\partial p} = -\frac{y}{p} + \frac{1-y}{1-p} = -\frac{1}{0.657} = -1.522$$

	chain	squared error
$\partial \mathcal{L}/\partial z$		-0.077
$\partial \mathcal{L}/\partial v_1$	$\partial \mathcal{L}/\partial z \times 0.731$	-0.056
$\partial \mathcal{L}/\partial w_{11}$	$\partial \mathcal{L}/\partial z \times 2 \times 0.197 \times 2$	-0.061

BACKPROPAGATION

Now with cross-entropy

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  $$\mathcal{L} = -y \log p - (1-y)\log(1-p)$$
  $$= -\log 0.657 = 0.420$$

- So only the first factor changes:

  $$\frac{\partial \mathcal{L}}{\partial p} = -\frac{y}{p} + \frac{1-y}{1-p} = -\frac{1}{0.657} = -1.522$$

- Times $\partial p / \partial z = p(1-p)$, it simplifies:

  $$\frac{\partial \mathcal{L}}{\partial z} = -y(1-p) + (1-y)p = p - y = -0.343$$

	chain	squared error
$\partial \mathcal{L}/\partial z$		-0.077
$\partial \mathcal{L}/\partial v_1$	$\partial \mathcal{L}/\partial z \times 0.731$	-0.056
$\partial \mathcal{L}/\partial w_{11}$	$\partial \mathcal{L}/\partial z \times 2 \times 0.197 \times 2$	-0.061

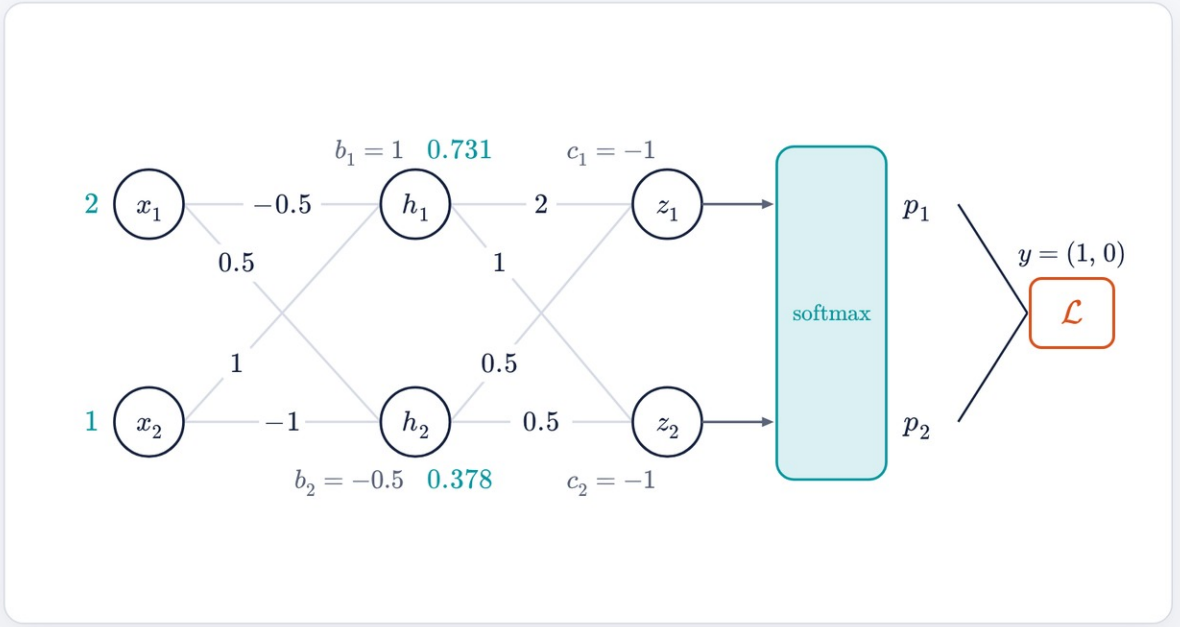
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  $$\mathcal{L} = -y \log p - (1-y)\log(1-p)$$
  $$= -\log 0.657 = 0.420$$
- So only the first factor changes:
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- Times $\partial p / \partial z = p(1-p)$, it simplifies:
  $$\frac{\partial \mathcal{L}}{\partial z} = -y(1-p) + (1-y)p = p - y = -0.343$$
- The rest of the chain is the same as before.

	chain	squared error	cross-entropy
$\partial \mathcal{L}/\partial z$		-0.077	-0.343
$\partial \mathcal{L}/\partial v_1$	$\partial \mathcal{L}/\partial z \times 0.731$	-0.056	-0.251
$\partial \mathcal{L}/\partial w_{11}$	$\partial \mathcal{L}/\partial z \times 2 \times 0.197 \times 2$	-0.061	-0.270

Two outputs

- Same input and first layer, so $h_1 = 0.731$, $h_2 = 0.378$. Now two outputs, a softmax and cross-entropy. The label is class 1: $\boldsymbol{y} = (1, 0)$.
  v_{kj} is the weight from h_j to z_k.

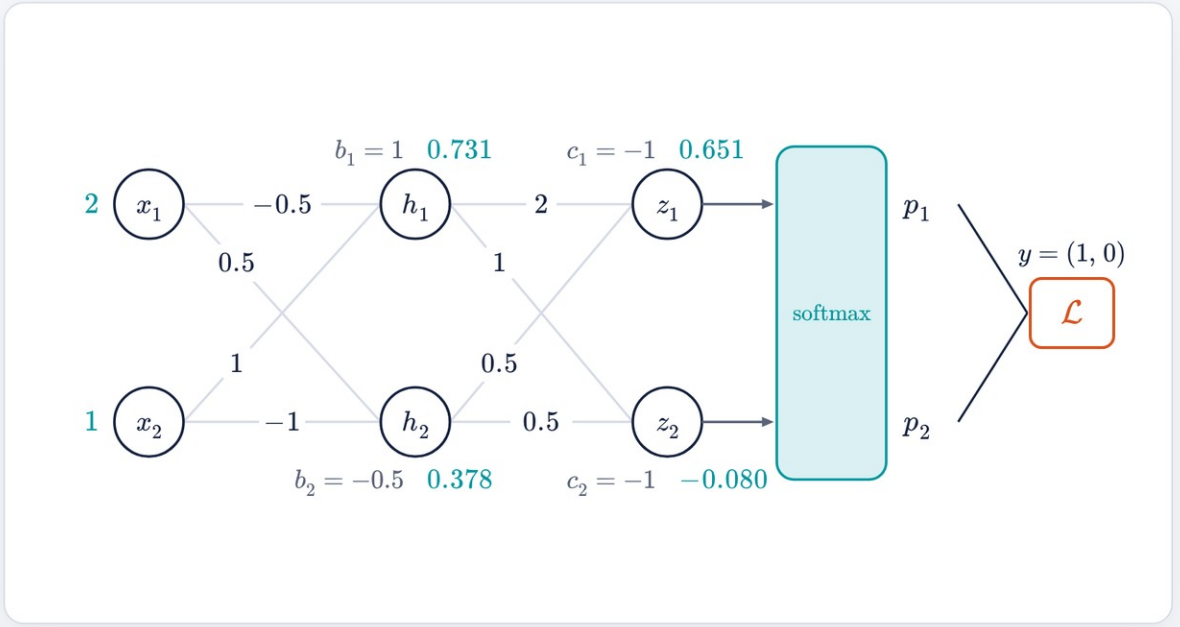


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  v_{kj} is the weight from h_j to z_k.
- The scores:

  $z_1 = 2 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = 0.651$

  $z_2 = 1 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = -0.080$



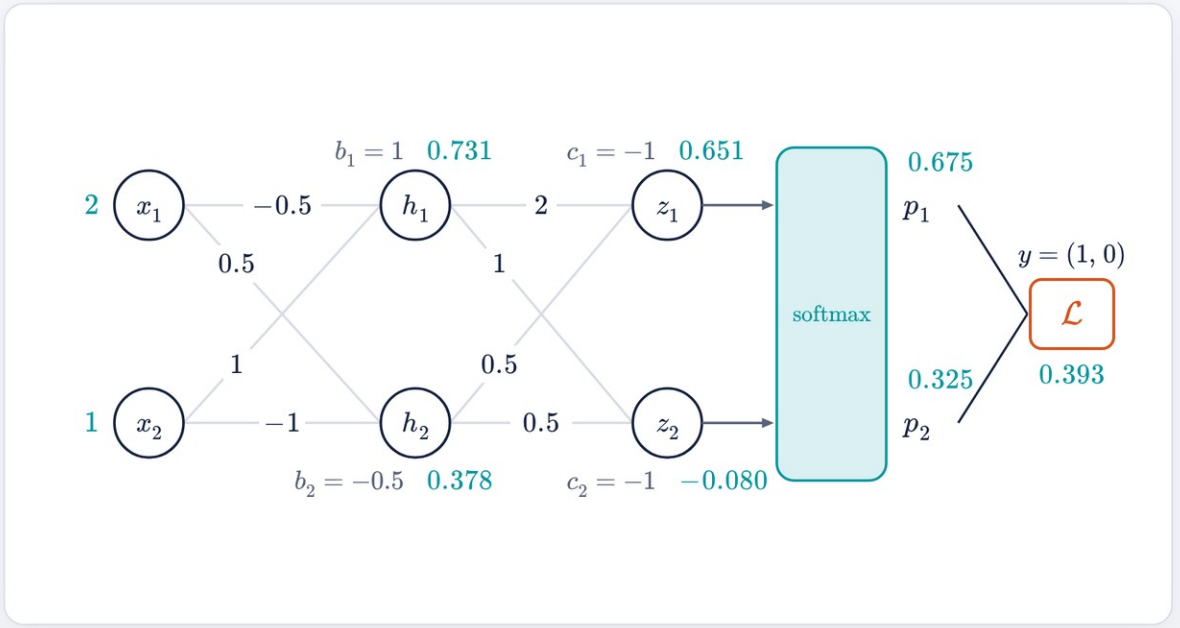
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  $$z_1 = 2 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = 0.651$$
  $$z_2 = 1 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = -0.080$$

- The softmax and the loss:

  $$p_1 = \frac{e^{0.651}}{e^{0.651} + e^{-0.080}} = \frac{1.917}{1.917 + 0.923} = 0.675$$
  $$p_2 = 0.325, \qquad \mathcal{L} = -\log 0.675 = 0.393$$



Two outputs

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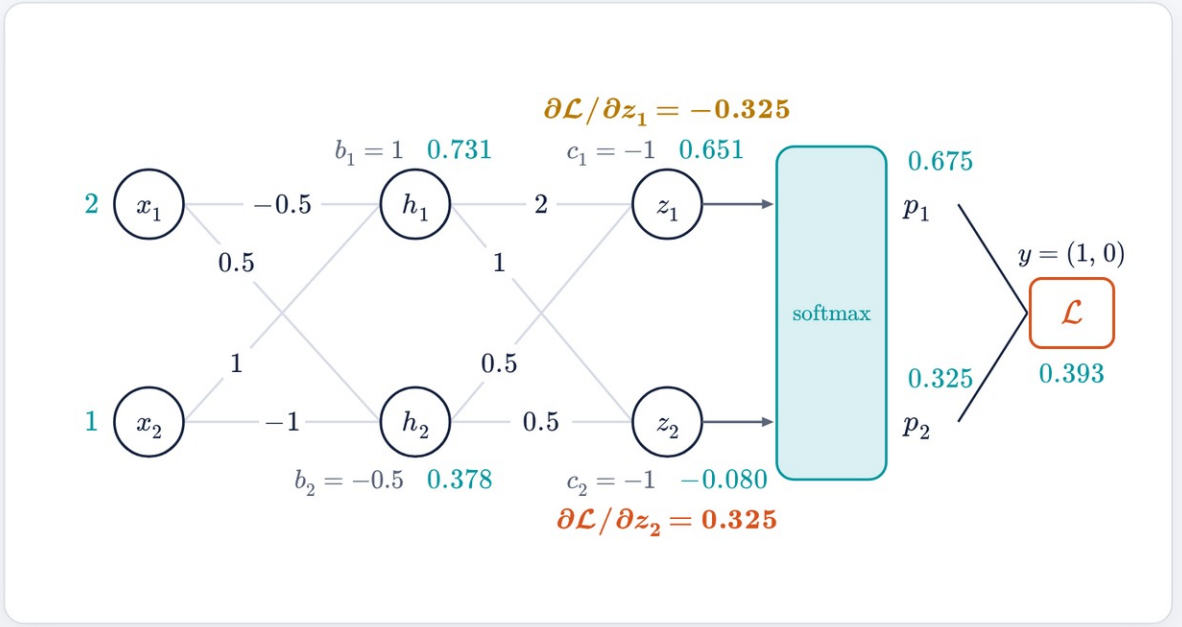
  $$z_1 = 2 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = 0.651$$
  $$z_2 = 1 \cdot 0.731 + 0.5 \cdot 0.378 - 1 = -0.080$$

- The softmax and the loss:

  $$p_1 = \frac{e^{0.651}}{e^{0.651} + e^{-0.080}} = \frac{1.917}{1.917 + 0.923} = 0.675$$
  $$p_2 = 0.325, \qquad \mathcal{L} = -\log 0.675 = 0.393$$

- At each output, as before, $\partial \mathcal{L} / \partial z_k = p_k - y_k$:

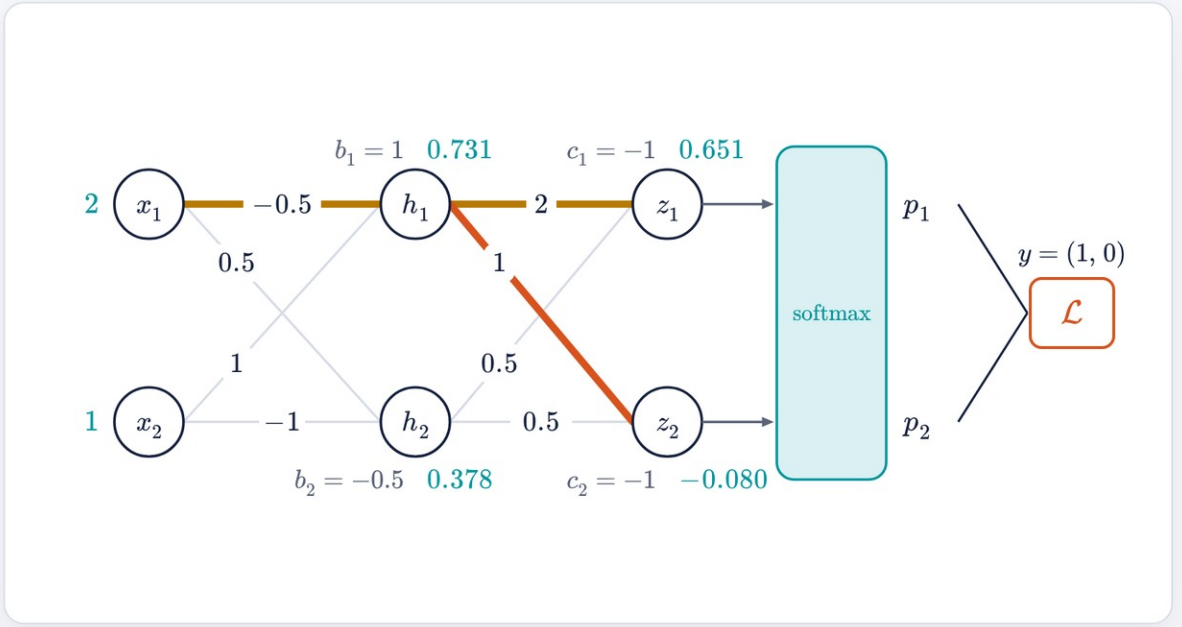
  $$\frac{\partial \mathcal{L}}{\partial z_1} = 0.675 - 1 = -0.325, \quad \frac{\partial \mathcal{L}}{\partial z_2} = 0.325 - 0 = 0.325$$



BACKPROPAGATION

Add the paths

- w_{11} changes h_1, and h_1 feeds both outputs. So w_{11} reaches the loss along two paths.

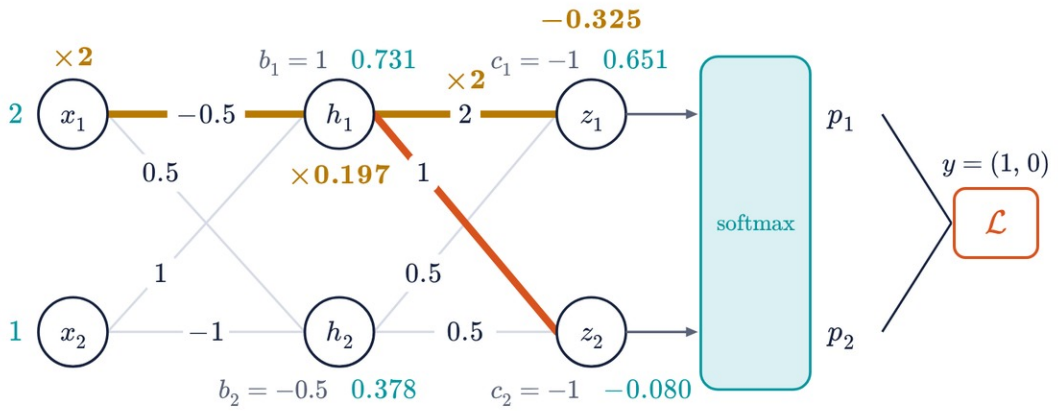


Add the paths

- w_{11} changes h_1, and h_1 feeds both outputs. So w_{11} reaches the loss along two paths.
- Through z_1:

  $(p_1 - y_1) \cdot v_{11} \cdot h_1(1 - h_1) \cdot x_1$

  $= -0.325 \times 2 \times 0.197 \times 2 = -0.256$



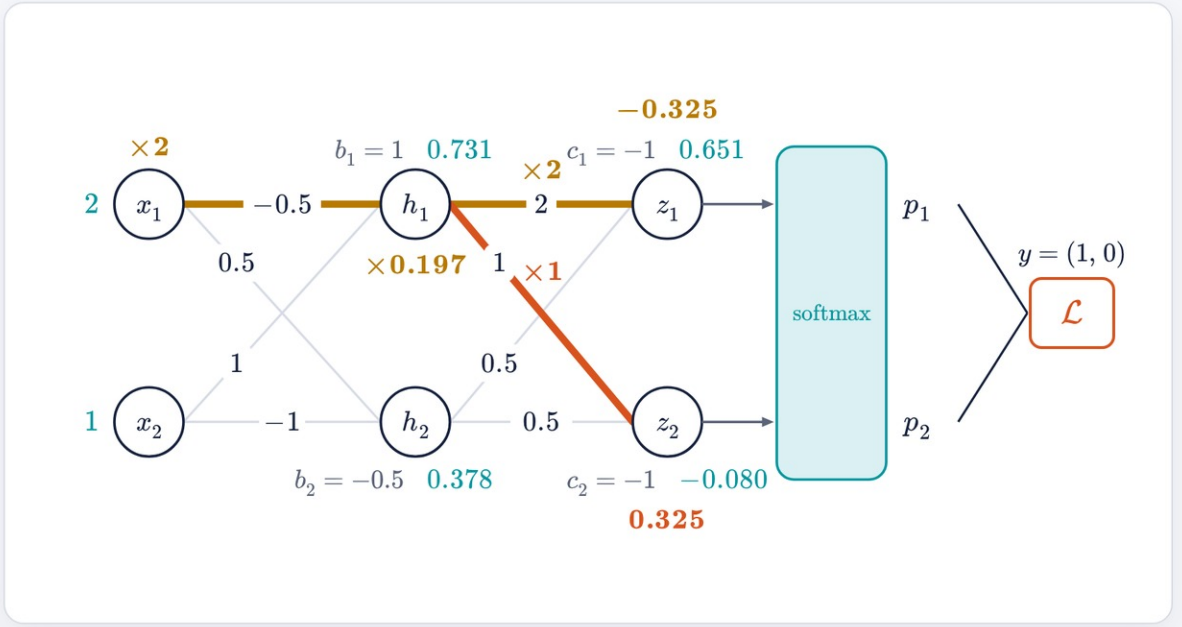
Add the paths

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  $$(p_1 - y_1) \cdot v_{11} \cdot h_1(1 - h_1) \cdot x_1$$
  $$= -0.325 \times 2 \times 0.197 \times 2 = -0.256$$

- Through z_2:

  $$(p_2 - y_2) \cdot v_{21} \cdot h_1(1 - h_1) \cdot x_1$$
  $$= 0.325 \times 1 \times 0.197 \times 2 = 0.128$$



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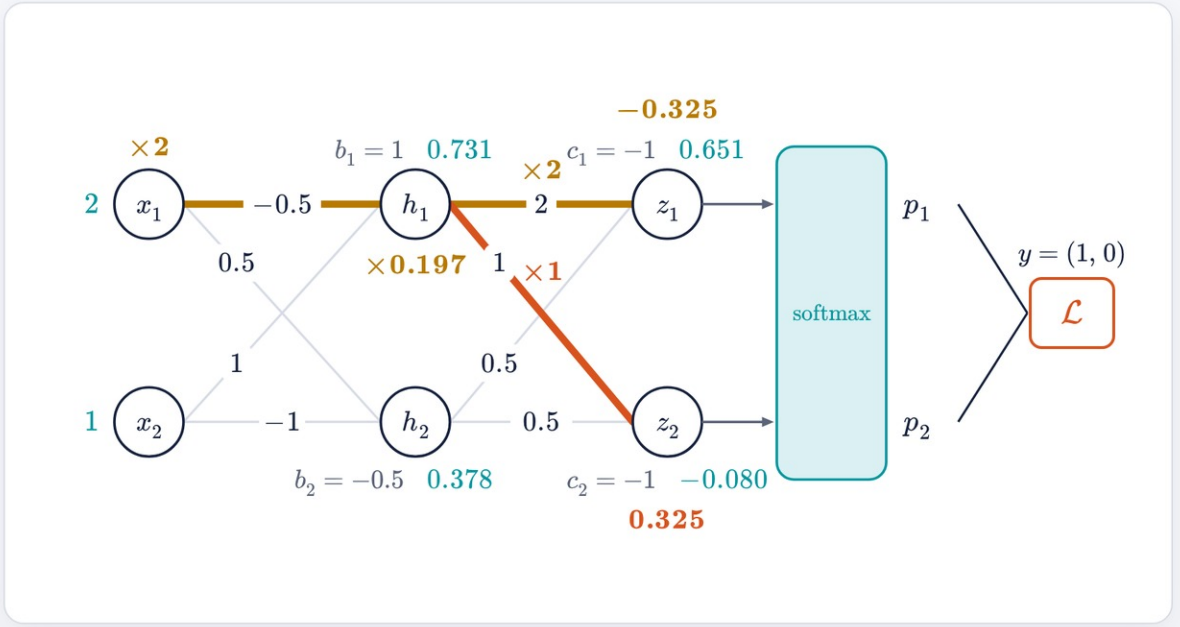
  $$(p_1 - y_1) \cdot v_{11} \cdot h_1(1 - h_1) \cdot x_1$$
  $$= -0.325 \times 2 \times 0.197 \times 2 = -0.256$$

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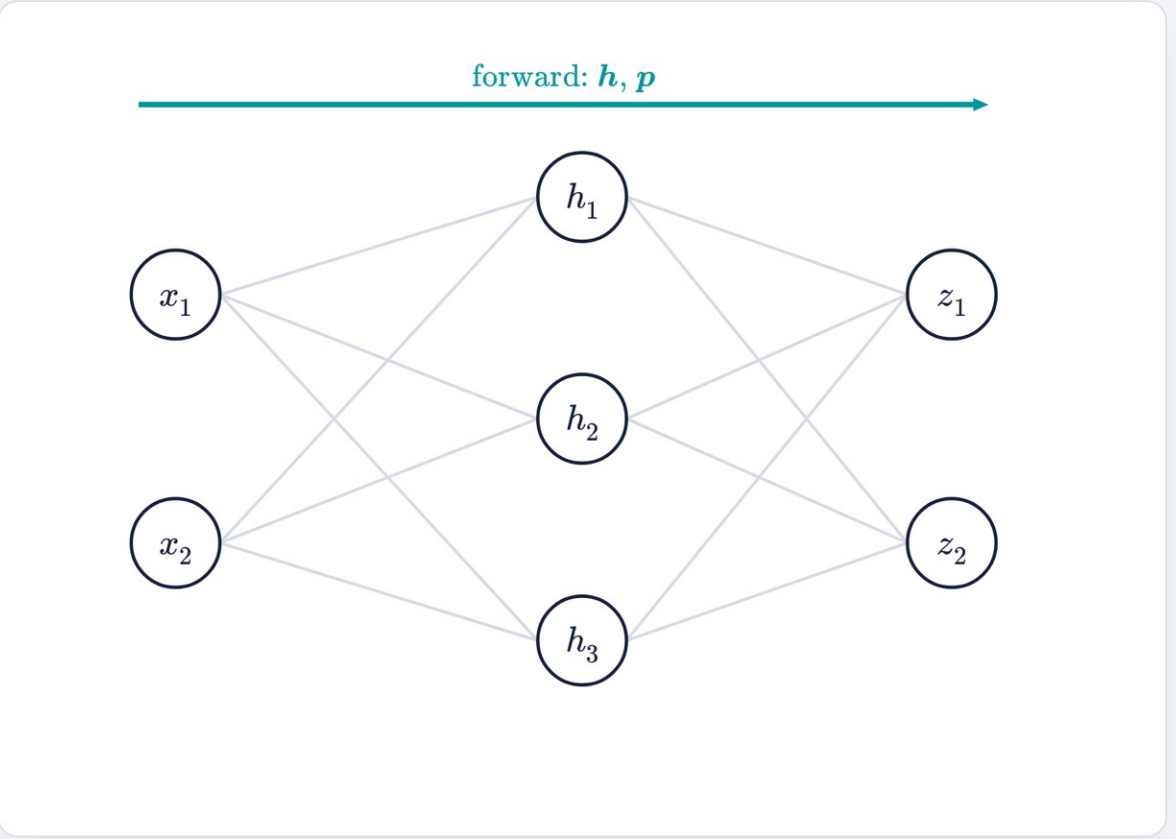
- Add them:

  $$\frac{\partial \mathcal{L}}{\partial w_{11}} = \sum_k (p_k - y_k)\, v_{k1} \cdot h_1(1 - h_1) \cdot x_1$$
  $$= -0.256 + 0.128 = -0.128$$



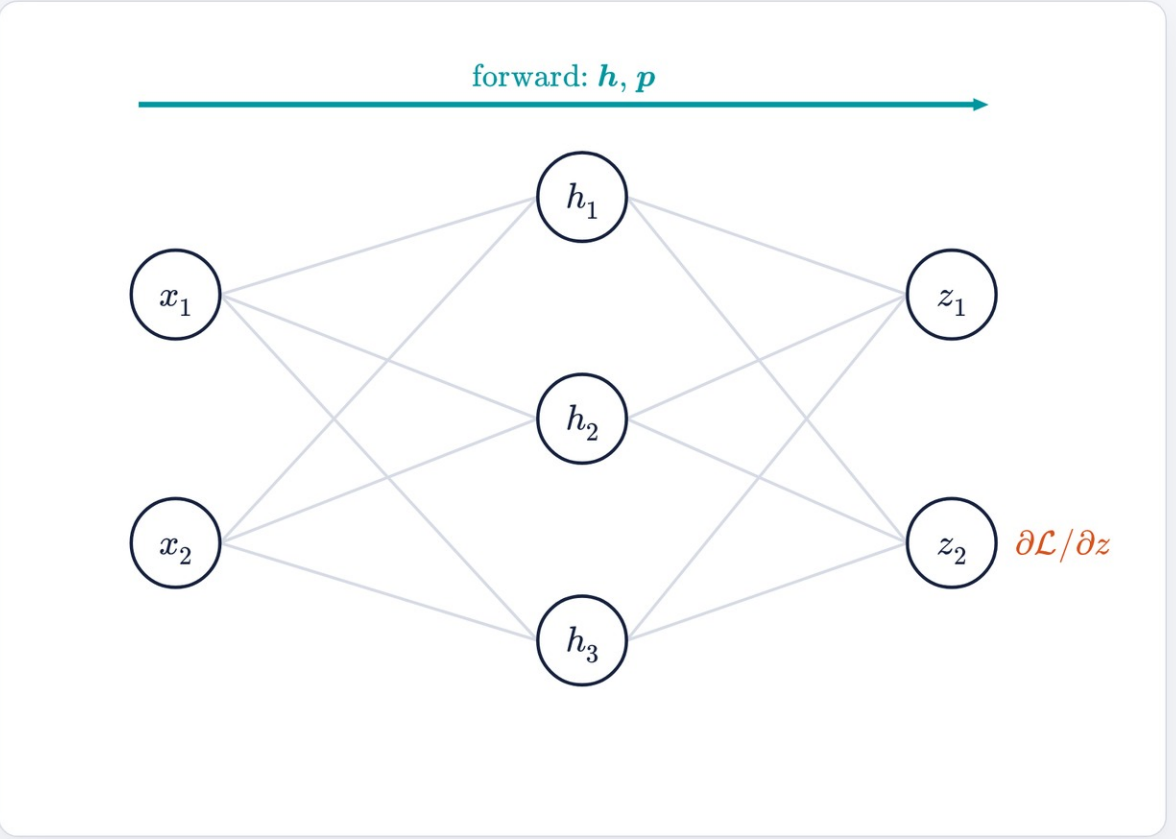
The recipe

1. **Forward pass**: compute and store $\boldsymbol{h}$ and $\boldsymbol{p}$.



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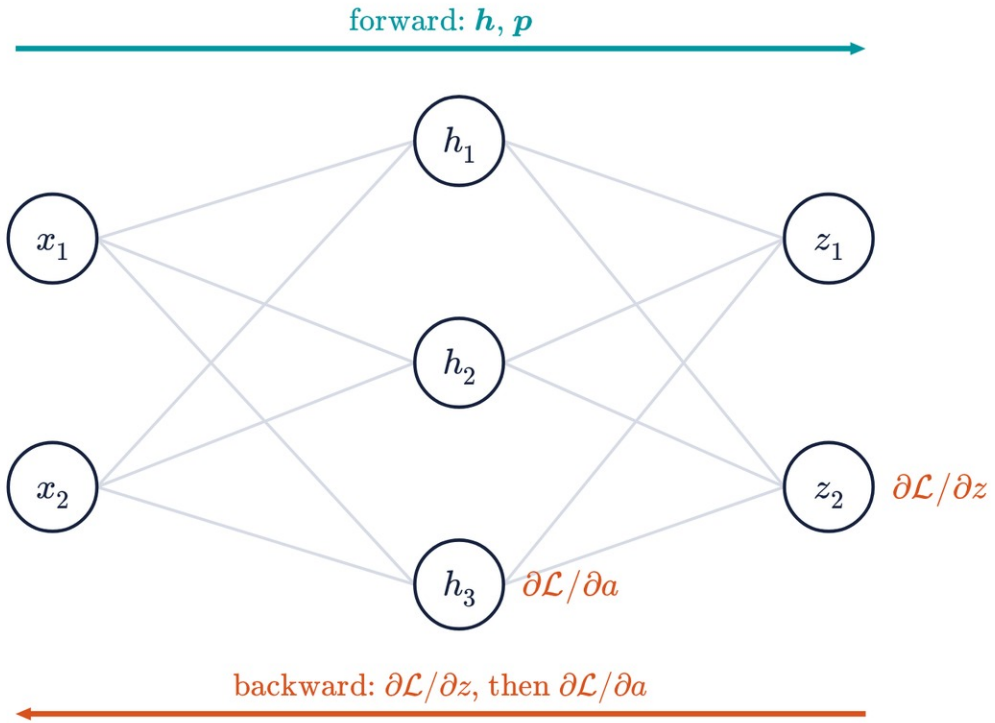


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$$\frac{\partial \mathcal{L}}{\partial a_j} = \left(\sum_k (p_k - y_k)\, v_{kj} \right) \cdot h_j (1 - h_j)$$

With another activation, use $\varphi'(a_j)$. With more layers, repeat, layer by layer.

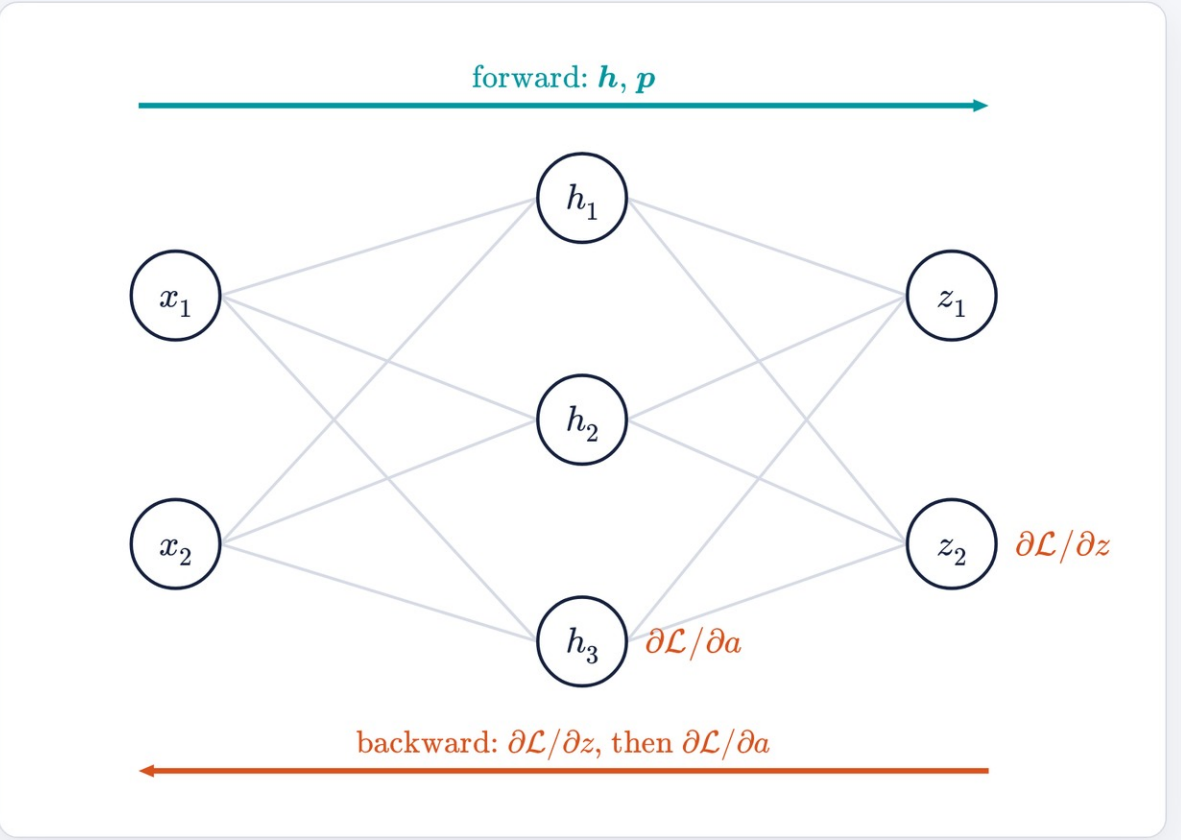


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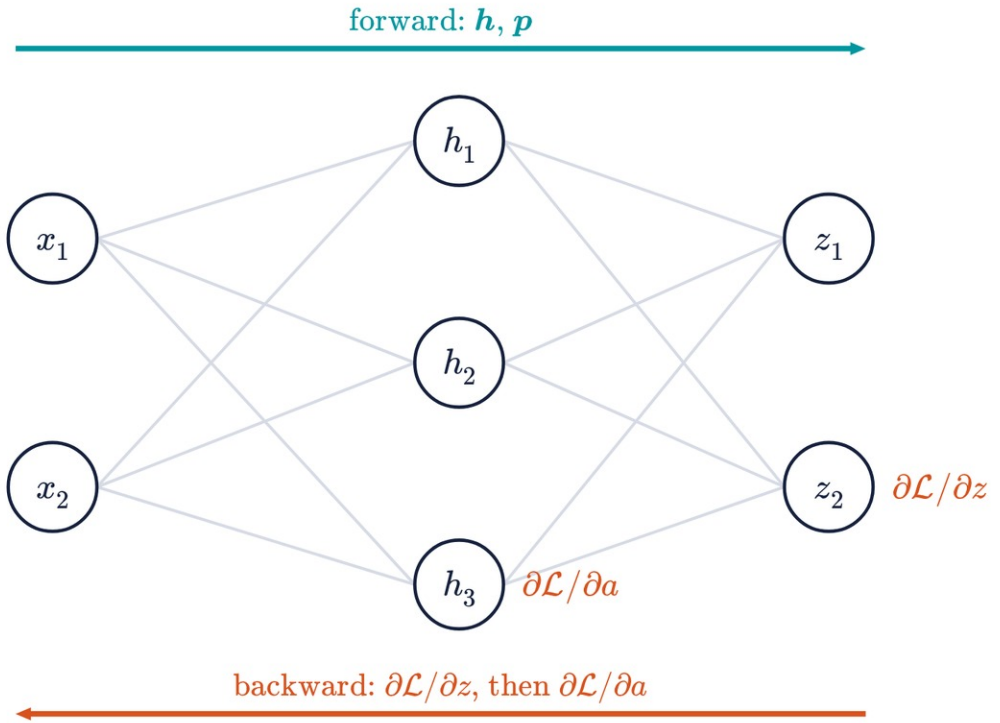


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5. **Update** every weight, $w \leftarrow w - \eta\, \partial \mathcal{L} / \partial w$, using the old weights throughout.



PUTTING IT TOGETHER

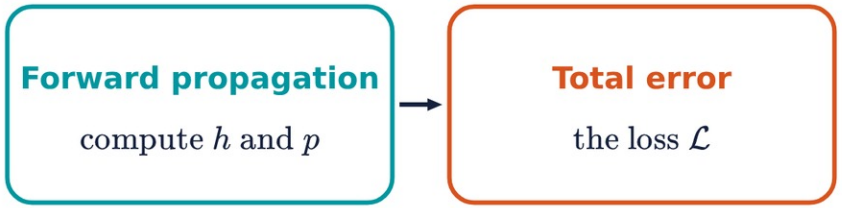
The training loop

Forward propagation

compute h and p

PUTTING IT TOGETHER

The training loop



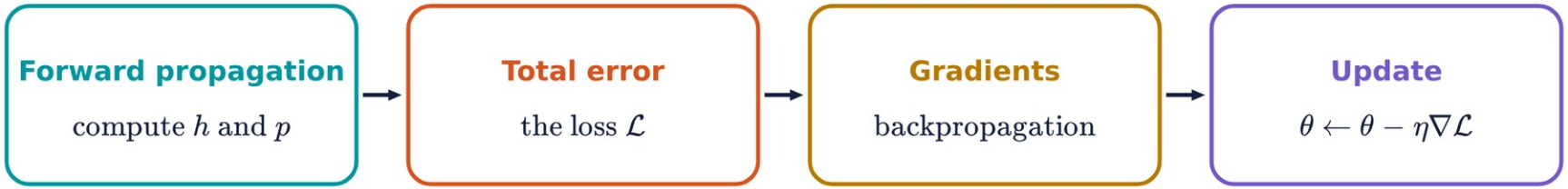
PUTTING IT TOGETHER

The training loop



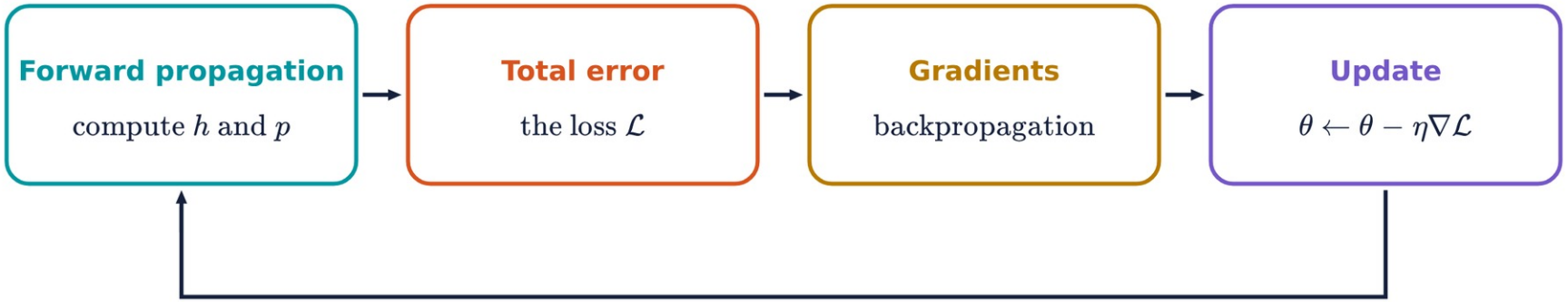
PUTTING IT TOGETHER

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PUTTING IT TOGETHER

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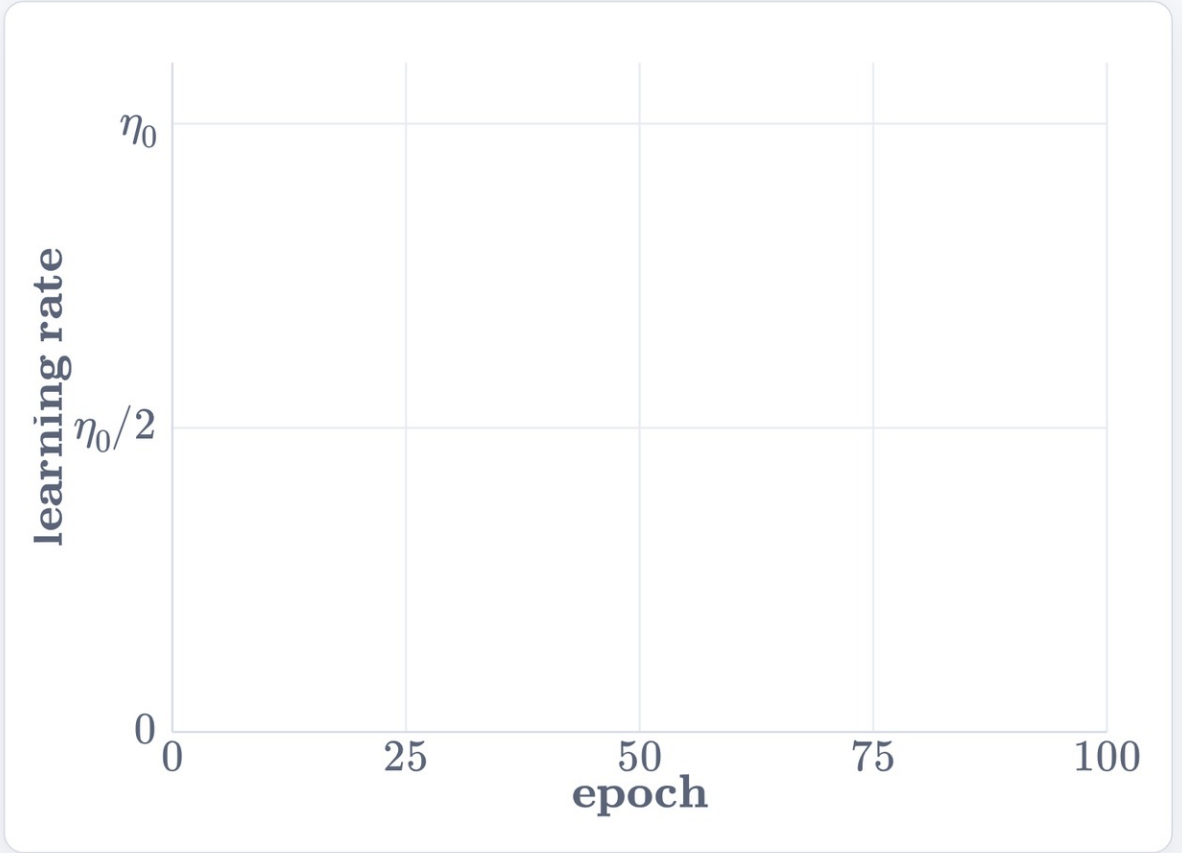
TRAINING IN PRACTICE

The learning rate

- The update:

  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla \mathcal{L}(\boldsymbol{\theta})$

  Too small: slow. Too large: it overshoots. (Last class.)



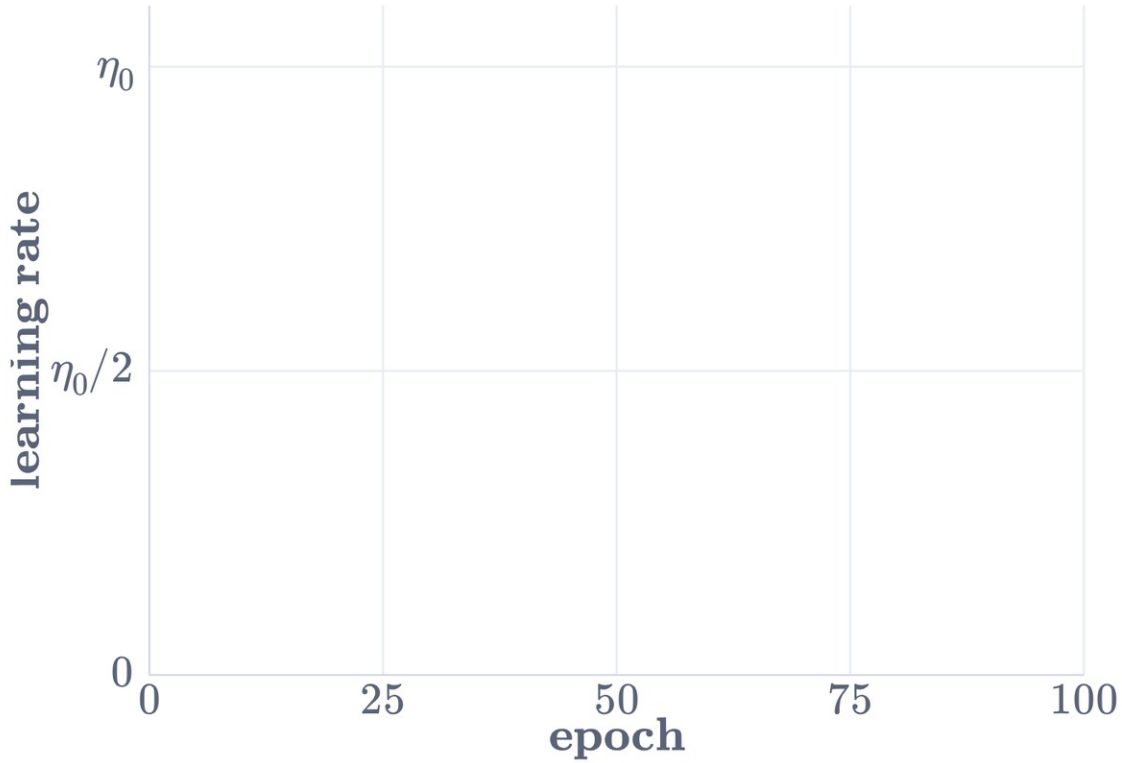
TRAINING IN PRACTICE

The learning rate

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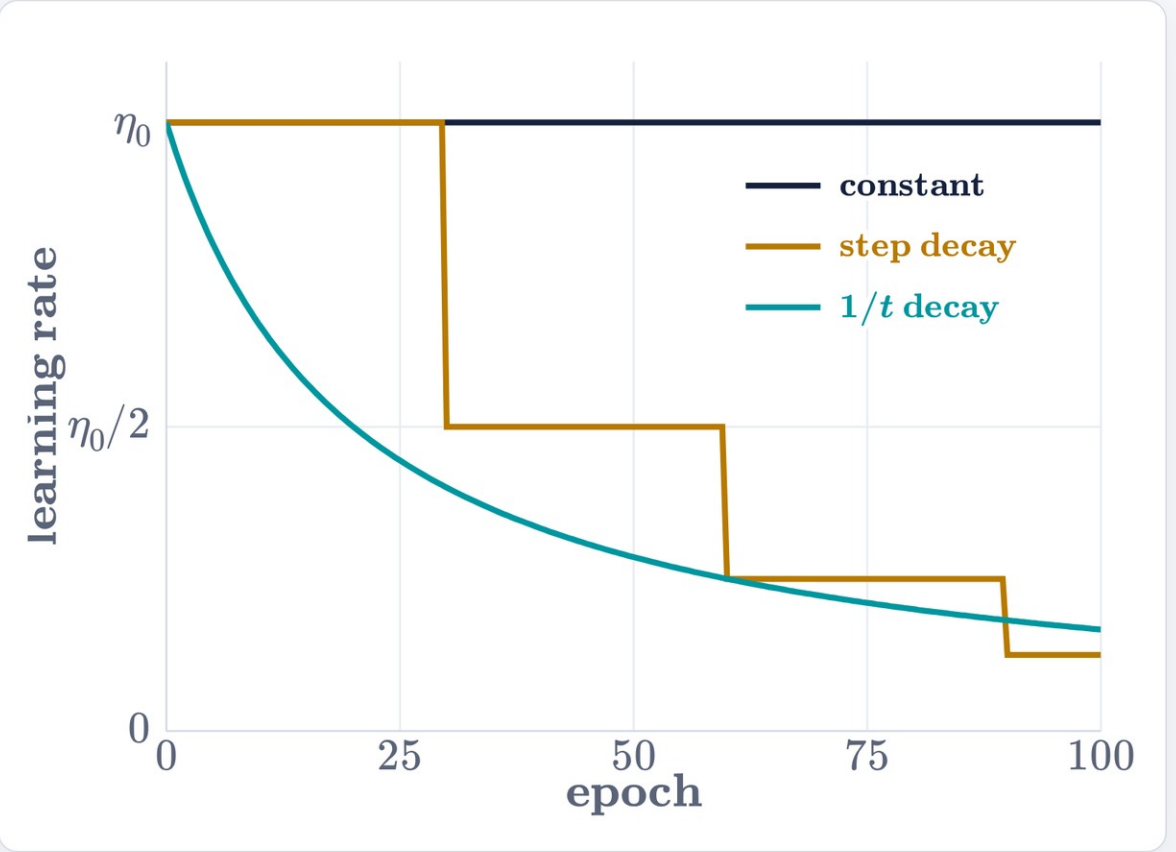
TRAINING IN PRACTICE

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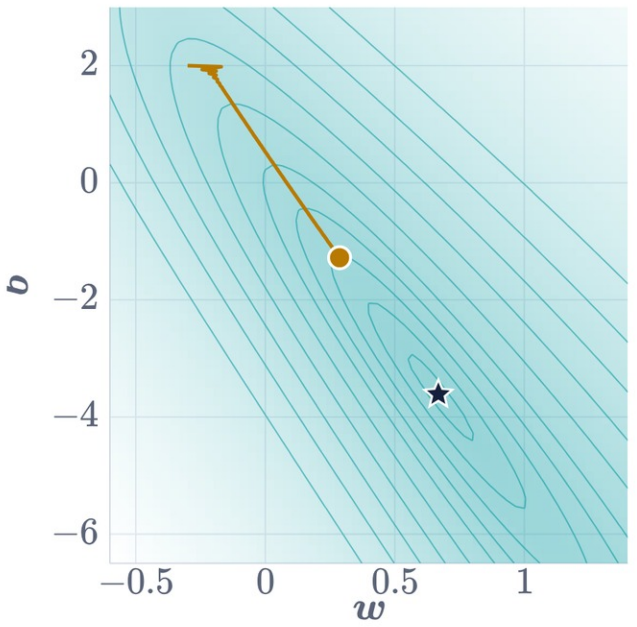
  Too small: slow. Too large: it overshoots. (Last class.)
- Large steps help early; small steps help near the end.
- A **learning-rate schedule** lowers η as training goes on.

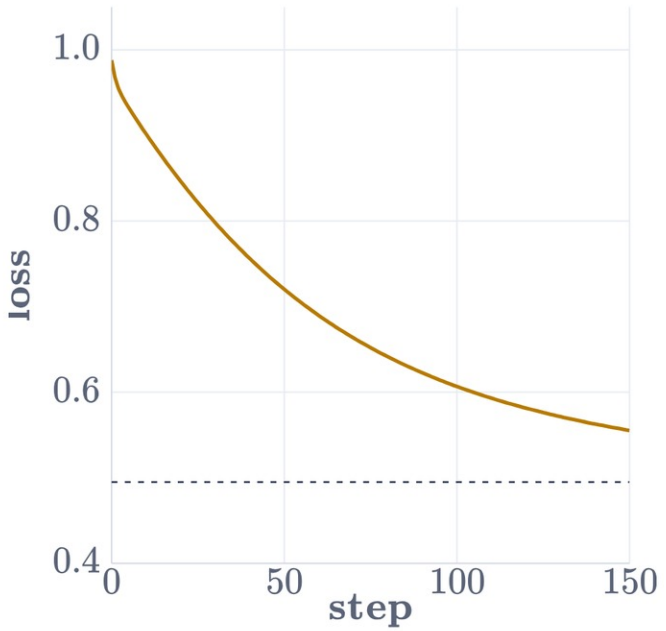


TRAINING IN PRACTICE

Momentum

- A long, narrow valley: plain gradient descent zigzags across it.





STEP
150

PLAIN LOSS
0.555

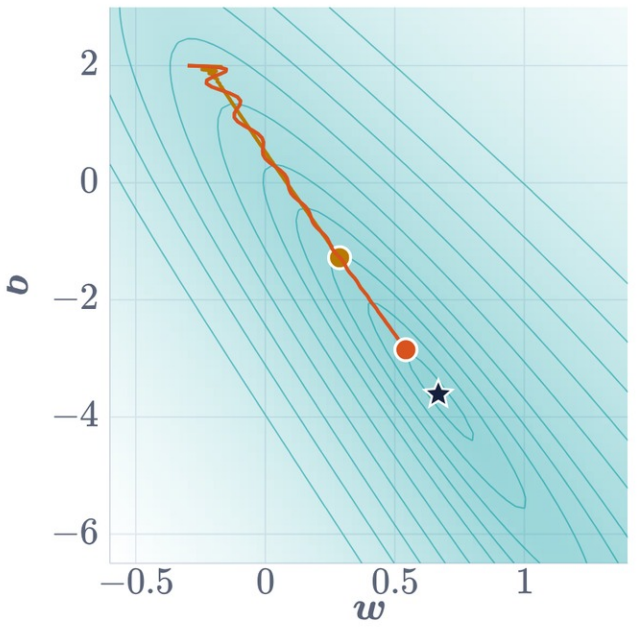
▶ step

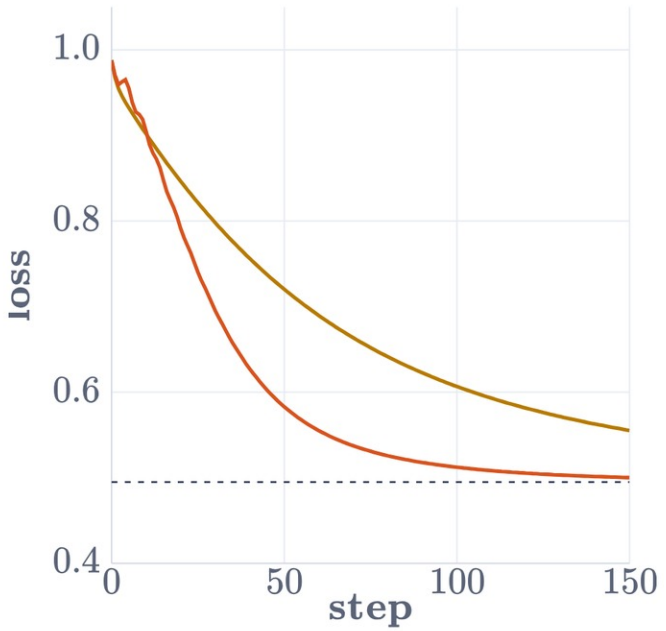
TRAINING IN PRACTICE

Momentum

- A long, narrow valley: plain gradient descent zigzags across it.
- Keep a running sum of past gradients:

  $\boldsymbol{v} \leftarrow \beta \boldsymbol{v} + \nabla \mathcal{L}, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{v}$





STEP	PLAIN LOSS	MOMENTUM LOSS
150	0.555	0.500

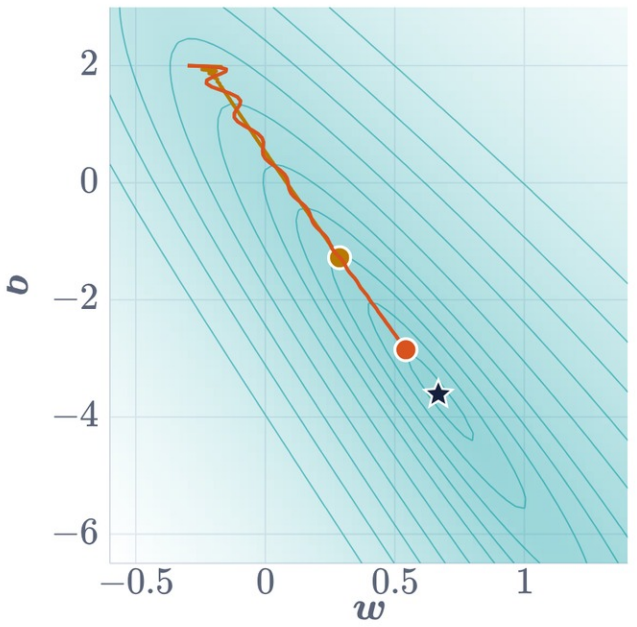
▶ step

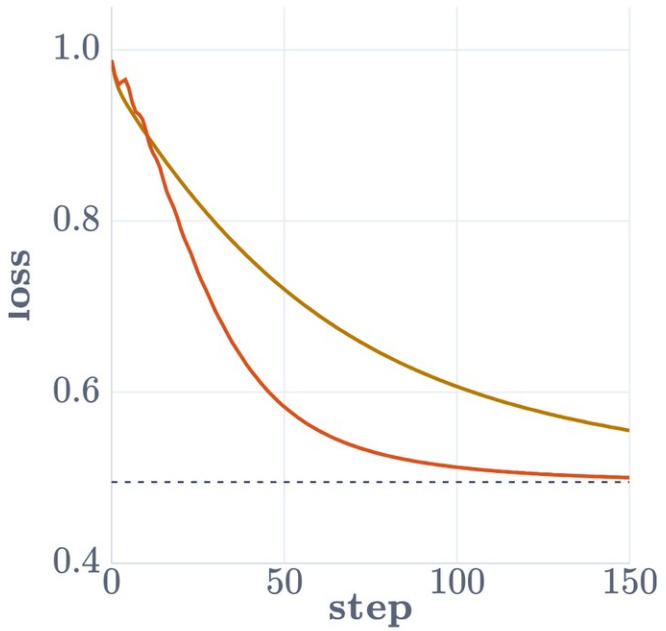
TRAINING IN PRACTICE

Momentum

- A long, narrow valley: plain gradient descent zigzags across it.
- Keep a running sum of past gradients:

  $\boldsymbol{v} \leftarrow \beta \boldsymbol{v} + \nabla \mathcal{L}, \quad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{v}$
- The zigzags cancel; the steady downhill part adds up.





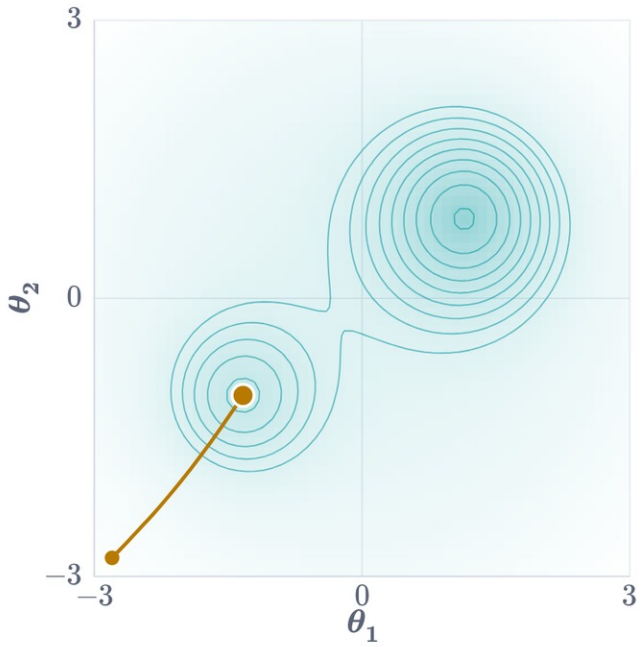
STEP	PLAIN LOSS	MOMENTUM LOSS
150	0.555	0.500

▶ step

TRAINING IN PRACTICE

Local minima

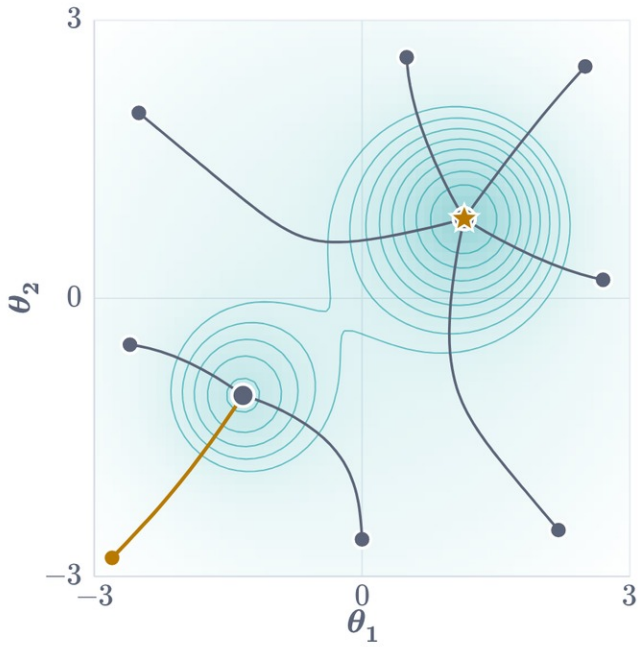
- Gradient descent stops in the valley it starts in.



TRAINING IN PRACTICE

Local minima

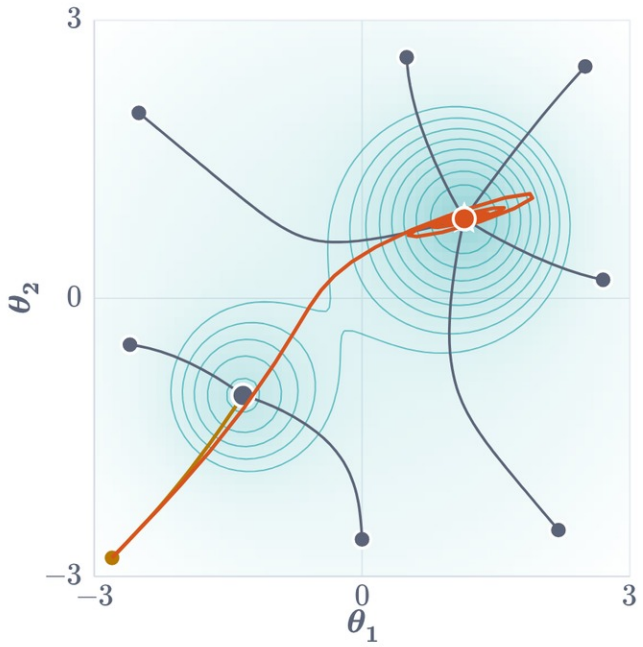
- Gradient descent stops in the valley it starts in.
- Restart from several starting points; keep the lowest loss.



TRAINING IN PRACTICE

Local minima

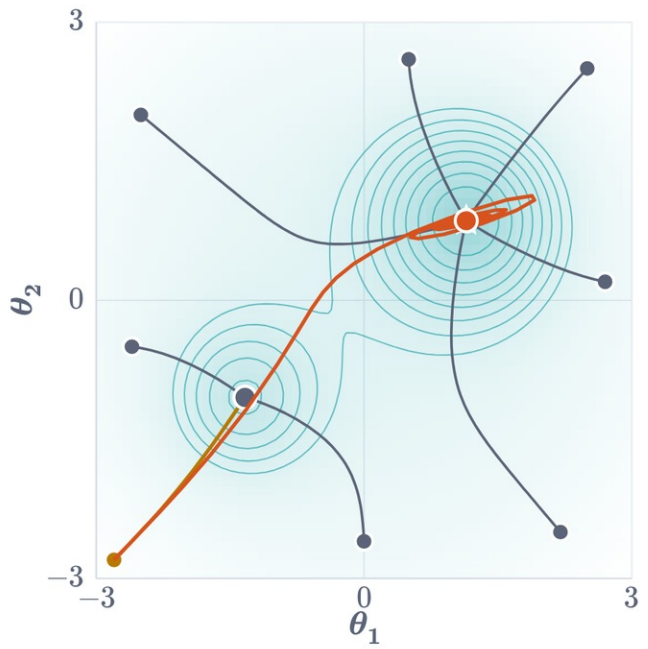
- Gradient descent stops in the valley it starts in.
- Restart from several starting points; keep the lowest loss.
- Momentum can roll through a shallow valley.

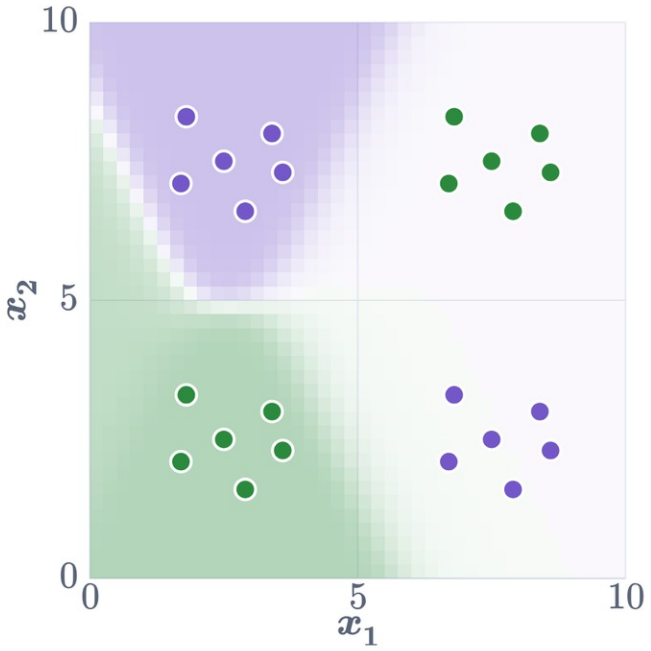


TRAINING IN PRACTICE

Local minima

- Gradient descent stops in the valley it starts in.
- Restart from several starting points; keep the lowest loss.
- Momentum can roll through a shallow valley.
- It happens on real networks: the XOR network from a different start stalls.



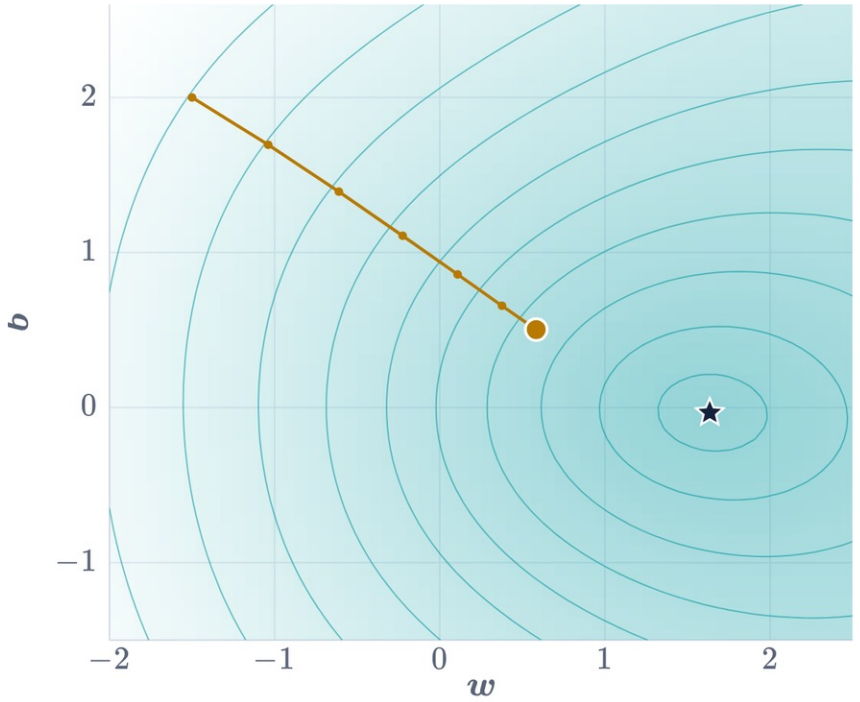


STEPS	LOSS	ACCURACY
4000	0.347	58%

TRAINING IN PRACTICE

How much data per step?

- **Batch**: the gradient from all the training points. One step per pass.



POINTS PER STEP
20

STEPS PER EPOCH
1

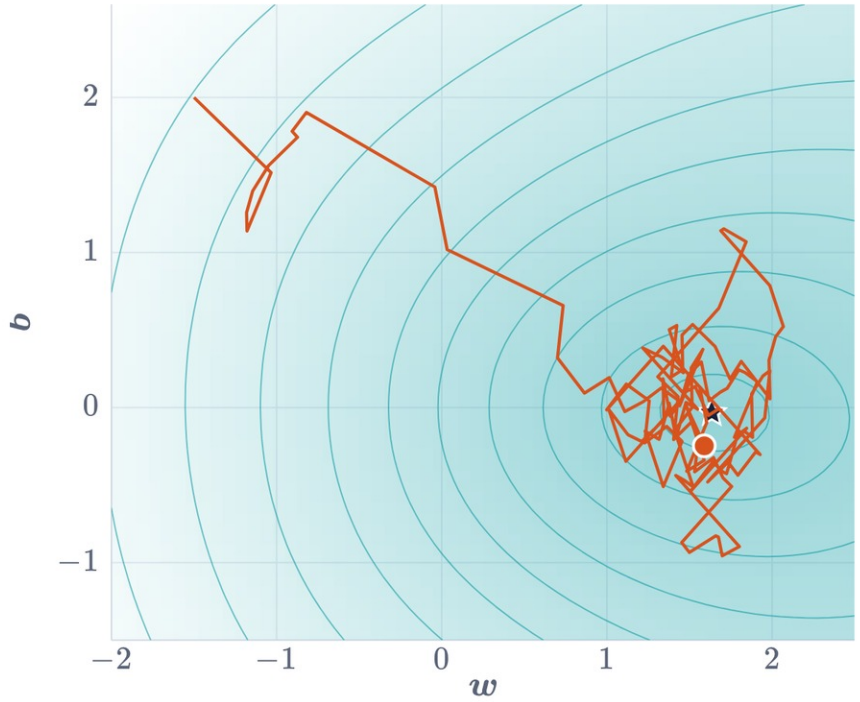
LOSS AFTER 6 EPOCHS
0.59

batch | stochastic | mini-batch | ▶

TRAINING IN PRACTICE

How much data per step?

- **Batch**: the gradient from all the training points. One step per pass.
- **Stochastic**: from one point at a time. Many cheap, noisy steps.



POINTS PER STEP
1

STEPS PER EPOCH
20

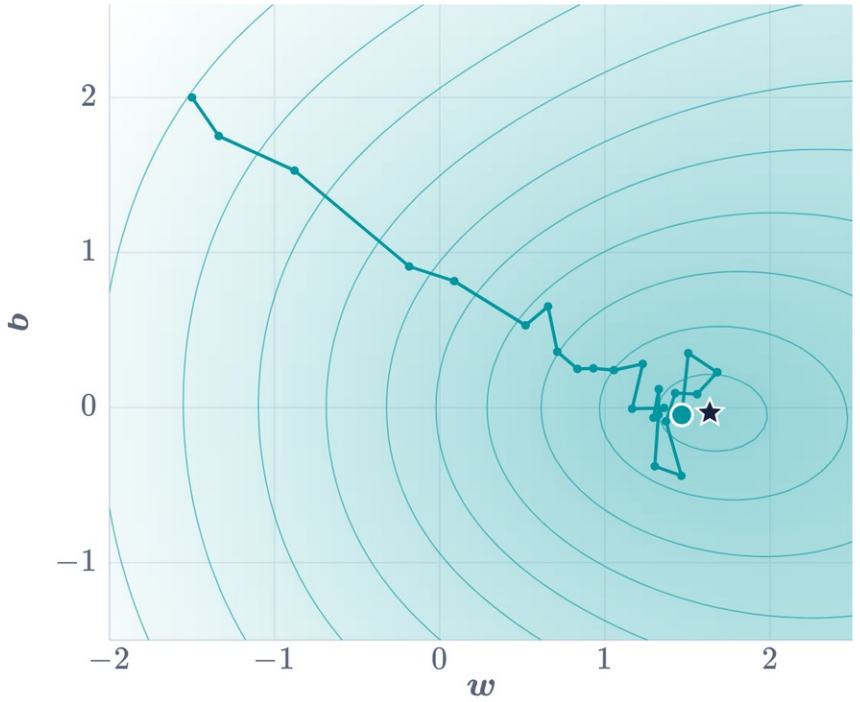
LOSS AFTER 6 EPOCHS
0.50

batch | stochastic | mini-batch | ▶

TRAINING IN PRACTICE

How much data per step?

- **Batch**: the gradient from all the training points. One step per pass.
- **Stochastic**: from one point at a time. Many cheap, noisy steps.
- **Mini-batch**: from a few points, here 5. In between.



POINTS PER STEP
5

STEPS PER EPOCH
4

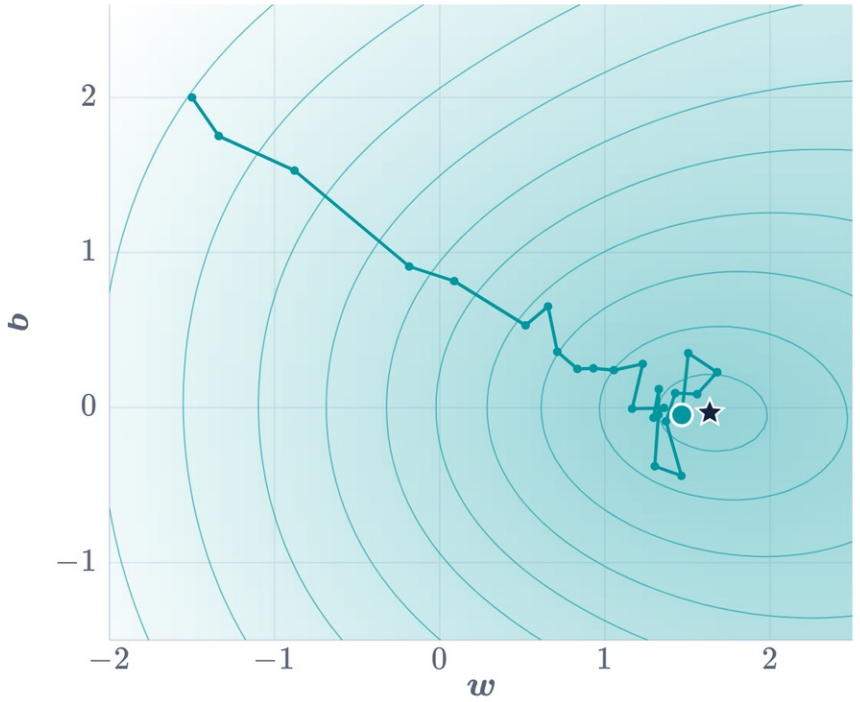
LOSS AFTER 6 EPOCHS
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batch | stochastic | mini-batch | ▶

TRAINING IN PRACTICE

How much data per step?

- **Batch**: the gradient from all the training points. One step per pass.
- **Stochastic**: from one point at a time. Many cheap, noisy steps.
- **Mini-batch**: from a few points, here 5. In between.
- An **epoch** is one pass through the training data.



POINTS PER STEP
5

STEPS PER EPOCH
4

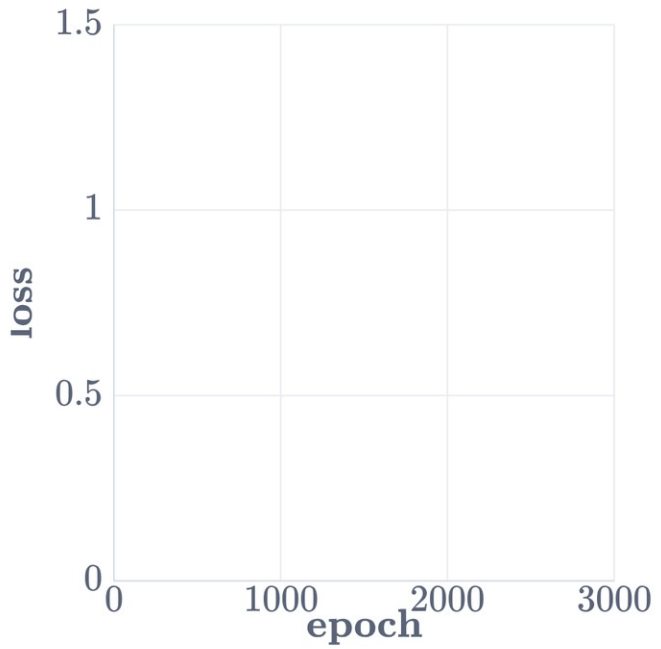
LOSS AFTER 6 EPOCHS
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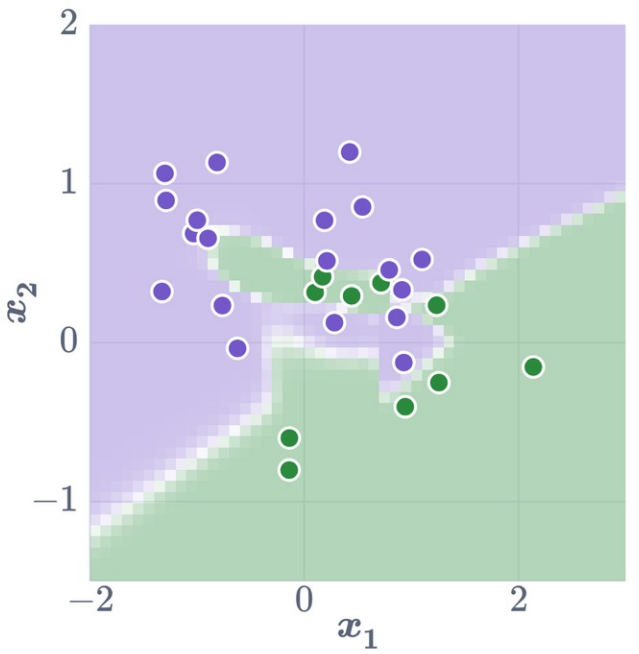
batch | stochastic | mini-batch | ▶

TRAINING IN PRACTICE

When to stop

- A maximum number of epochs; the weights stop changing; the loss is low enough.



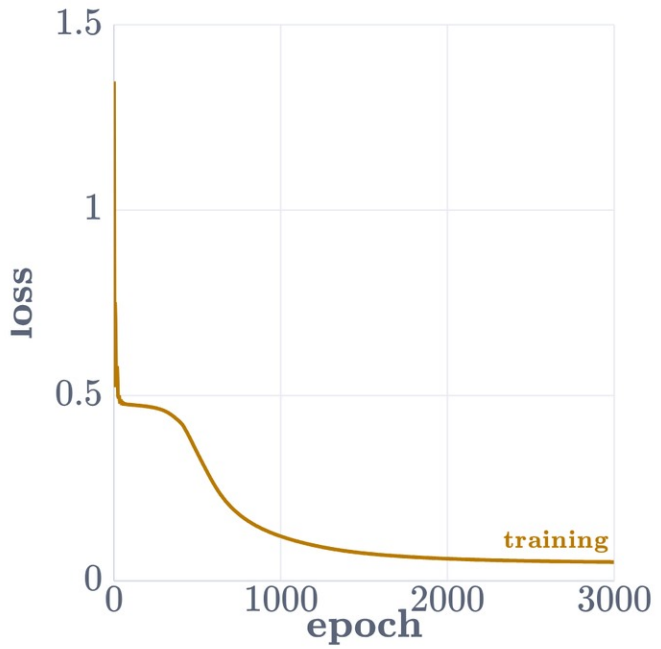


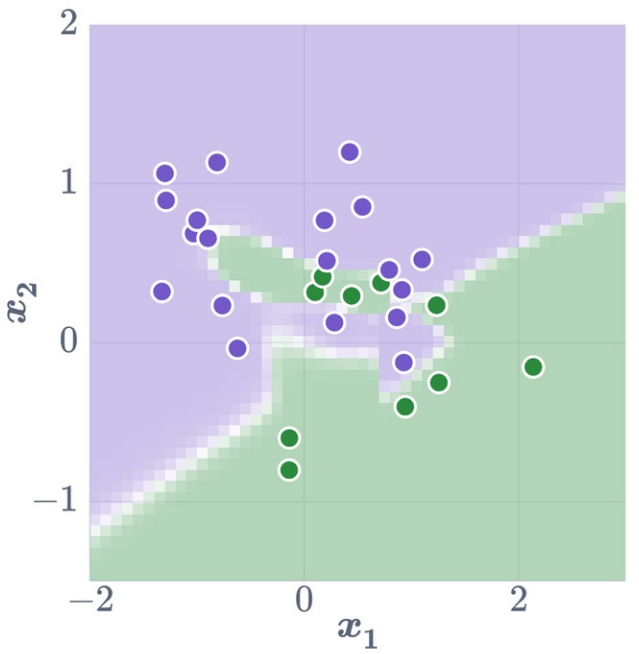
EPOCH	TRAINING ACCURACY	VALIDATION ACCURACY
3000	93%	77%

TRAINING IN PRACTICE

When to stop

- A maximum number of epochs; the weights stop changing; the loss is low enough.
- The training loss keeps going down.



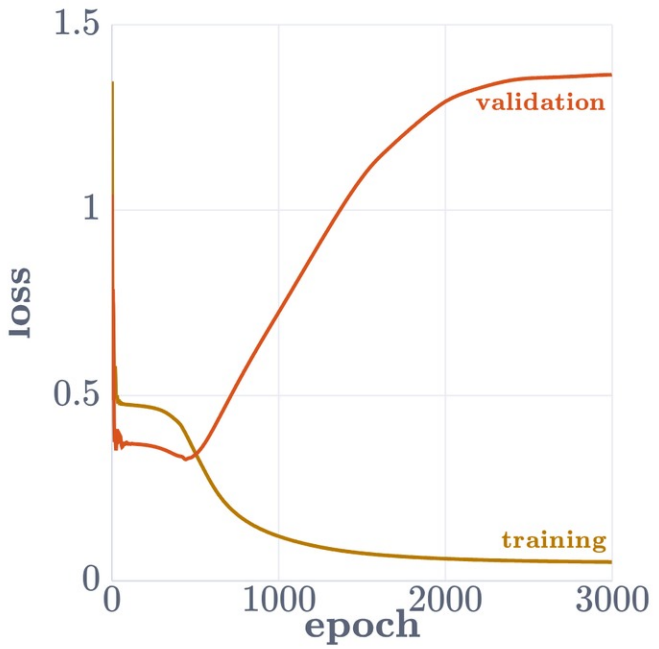


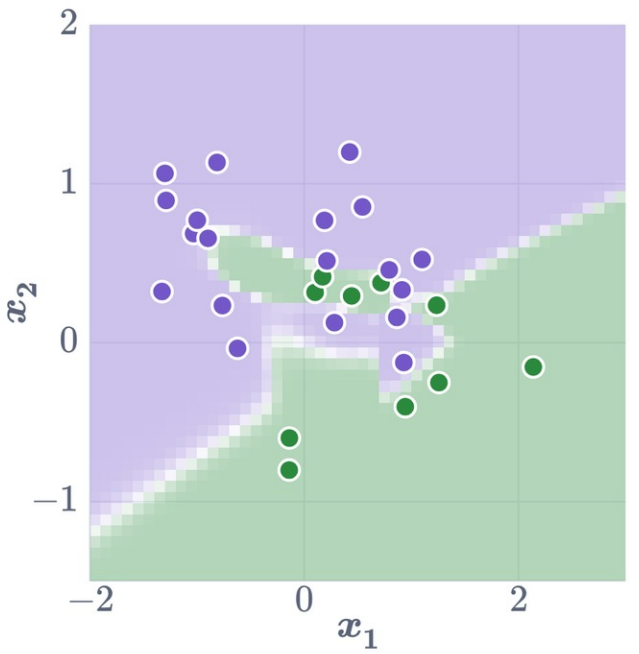
EPOCH	TRAINING ACCURACY	VALIDATION ACCURACY
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TRAINING IN PRACTICE

When to stop

- A maximum number of epochs; the weights stop changing; the loss is low enough.
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- The validation loss goes down, then up: the network starts to overfit.





EPOCH
3000

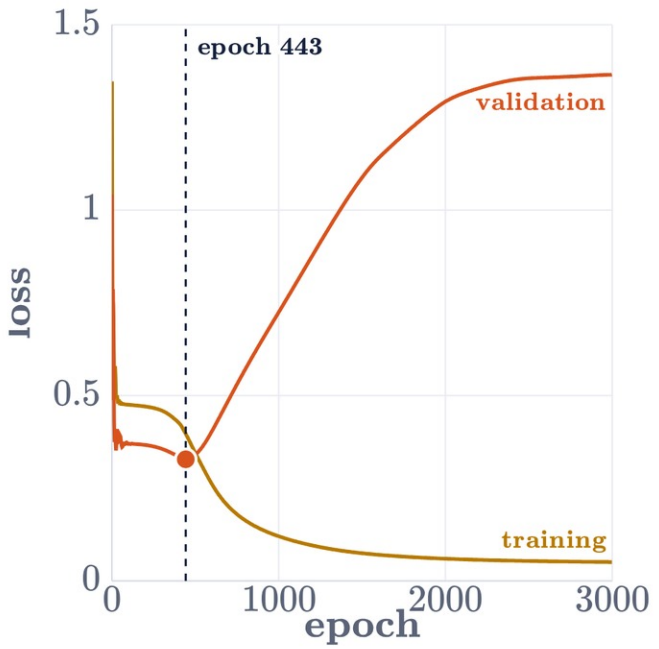
TRAINING ACCURACY
93%

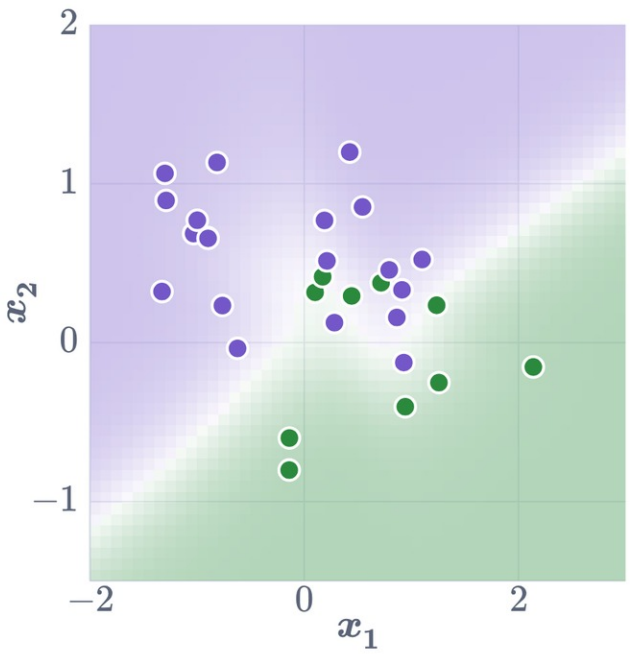
VALIDATION ACCURACY
77%

TRAINING IN PRACTICE

When to stop

- A maximum number of epochs; the weights stop changing; the loss is low enough.
- The training loss keeps going down.
- The validation loss goes down, then up: the network starts to overfit.
- **Early stopping**: keep the weights from the epoch with the lowest validation loss.



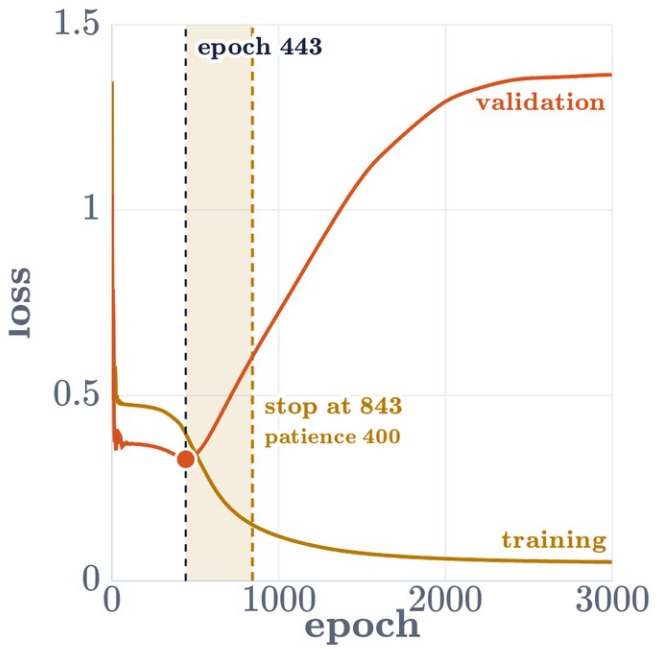


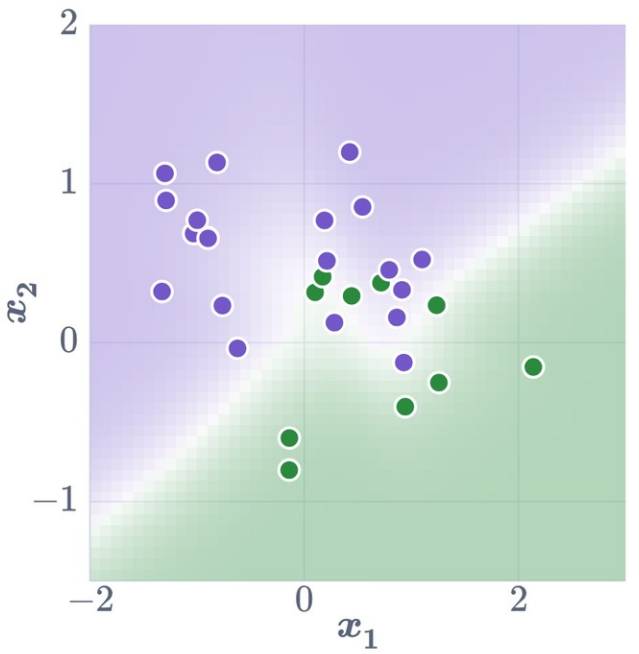
EPOCH	TRAINING ACCURACY	VALIDATION ACCURACY
443	80%	89%

TRAINING IN PRACTICE

When to stop

- A maximum number of epochs; the weights stop changing; the loss is low enough.
- The training loss keeps going down.
- The validation loss goes down, then up: the network starts to overfit.
- **Early stopping**: keep the weights from the epoch with the lowest validation loss.
- **Patience**: stop once the validation loss has not improved for a set number of epochs.





EPOCH	TRAINING ACCURACY	VALIDATION ACCURACY
443	80%	89%

SUMMARY

Neural networks: characteristics

Strengths

- Very flexible: one hidden layer with enough units can approximate any continuous function on a bounded region.
- They learn their own features, so less hand-made feature design.
- They improve with more data and computing power, and handle many outputs at once.
- Irrelevant inputs can be ignored: training can drive their weights to zero.

SUMMARY

Neural networks: characteristics

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Weaknesses

- Non-convex training: the answer depends on the start, with no guarantee of the best weights.
- Many choices to tune: layers, units, η, optimizer, epochs.
- They need a lot of data, overfit small data sets, and are hard to interpret.
- Training is time-consuming: many passes over the data, and many runs to tune the choices above.

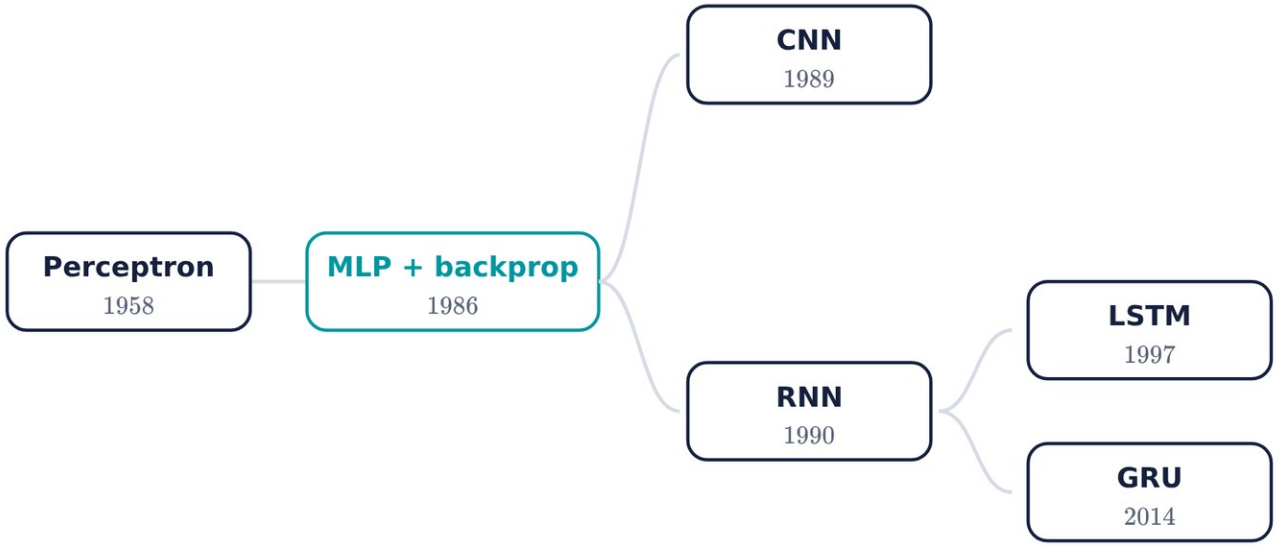
WHERE THIS GOES

The family tree



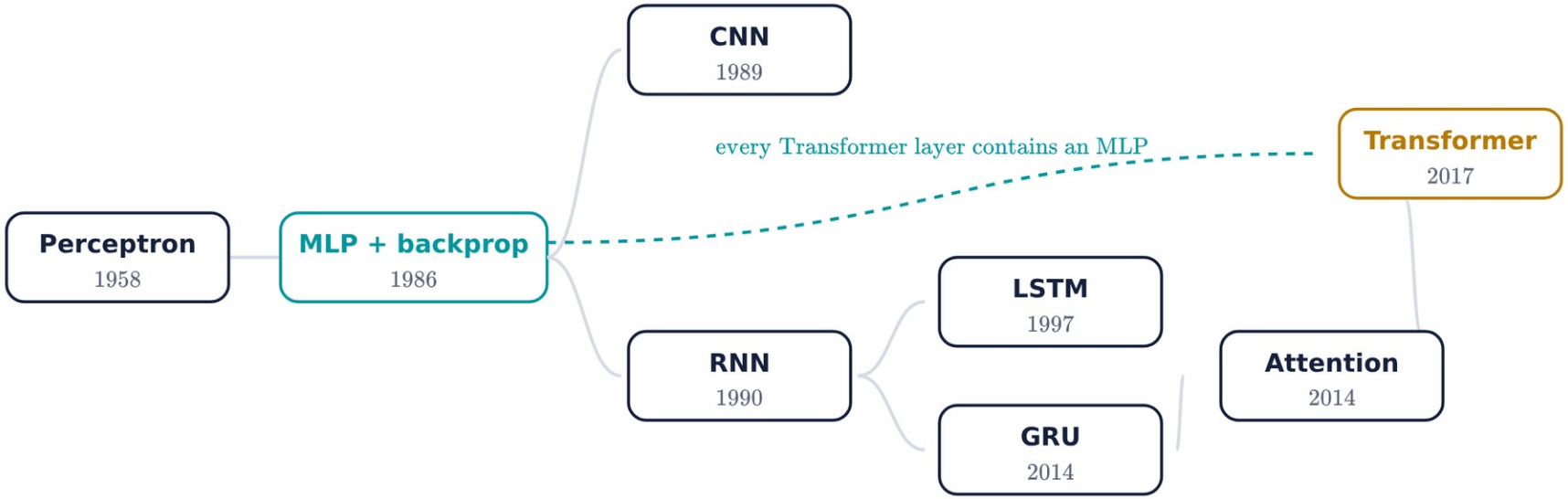
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SUMMARY

Takeaways

Backpropagation
The chain rule, from the output back. At the output, $p_k - y_k$; a hidden unit adds up the paths through every output it feeds. Each weight's gradient is the error at its end times the value at its start.

Optimizers
Learning-rate schedules and momentum.

Batches
Batch, mini-batch, stochastic. An epoch is one pass through the data.

Stopping
Watch the validation loss; stop early.

Thank you.

Questions?